

HW01 – QA/QC Jobs, Defects and Physical Product

Requirement 1 – QA/QC Job Market 2026+ (40 points)

1.1 Search scope

- **Submission date:** 27/09/2026
- **60-day window:** 30/07/2026–28/09/2026
- **Number of postings:** 10
- **Postings requiring AI/LLM/AI-assisted automation skills:** 6 (JP01, JP02, JP04, JP06, JP08, JP09)
- **Screenshot folder:** R1_Job_Postings/

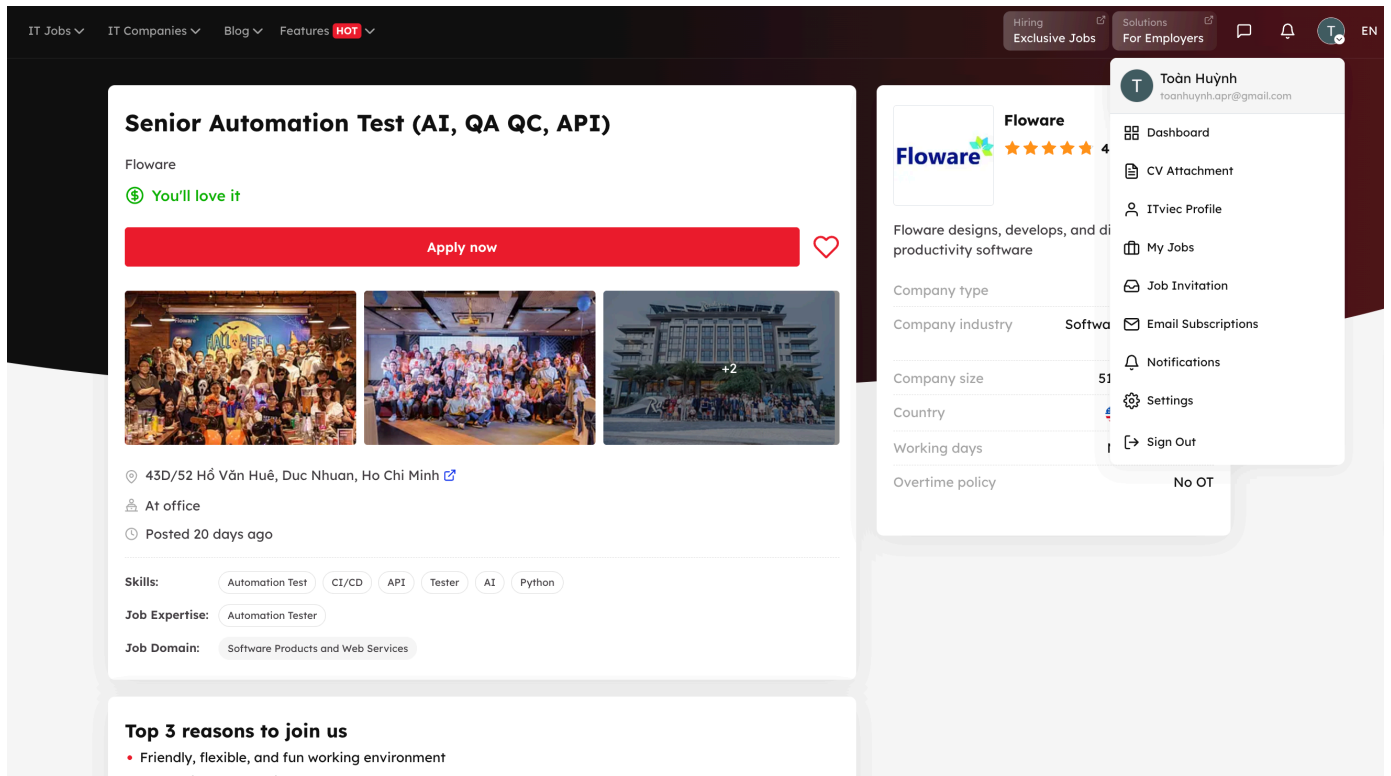
1.2 Summary of collected postings

ID	Position	Company	Platform	Posting date	AI/LLM/AI-automation requirement	Salary	Evidence
JP01	Senior Automation Test (AI, QA QC, API)	Floware	ITviec	04/09/2026	Yes – AI tools (Claude, Cursor, Copilot, ChatGPT) for test design, automation, debugging and quality analysis	Not disclosed	JP01.png
JP02	AI Quality Assurance Tester (Data Testing)	SCC Vietnam	LinkedIn	3 weeks ago (as of 23/09/2026)	Yes – AI testing, conversational interfaces, Azure AI, EU AI Act	Not disclosed	JP02.png
JP03	Quality Engineering	DXC Technology	DXC Careers (Workday)	08/09/2026	No	Not disclosed	JP03.png
JP04	Senior QA Engineer (AI-Augmented Quality Engineering)	Ins Enco	LinkedIn	1 week ago (as of 23/09/2026)	Yes – GPT/Claude test generation, Mabl/Testim, AppliTools/Percy, Postbot	30–40 million VND gross/month	JP04.png
JP05	Manual Tester (QA QC)	Công ty cổ phần Tập đoàn Công nghệ Quảng Ích (QIG)	ITviec	23/09/2026	No	500–1,200 USD/month	JP05.png
JP06	Middle QA Automation Engineer (Playwright, Selenium)	FPT Digital	ITviec	15/09/2026	Yes – testing AI/LLM features (chatbots, RAG, hallucination detection); GenAI assistants for test generation	800–1,500 USD/month	JP06.png
JP07	QA Engineer (Tester, QA QC, English) Up to \$1500	Saritasa	ITviec	16/09/2026	No (AI tools listed only as a preferred skill)	1,000–1,500 USD/month	JP07.png
JP08	[Hanoi] Fullstack QA Engineer Lead (Manual, Auto, AI)	Money Forward Vietnam	ITviec	14/09/2026	Yes – GenAI tools (Cursor, GitHub Copilot, LLM assistants), AI-augmented testing, testing AI features	Not disclosed	JP08.png
JP09	QA Engineer (Claude Code, Python, React, Manual Tester)	Brarista	ITviec	07/09/2026	Yes – Claude Code for test matrices/regression suites, verifying AI-suggested tags	800–1,000 USD/month	JP09.png
JP10	Software Quality Analyst (QA, Tester)	MiTek Vietnam	ITviec	21/08/2026	No	Not disclosed	JP10.png

1.3 Detailed posting records

JP01 – Senior Automation Test (AI, QA QC, API), Floware

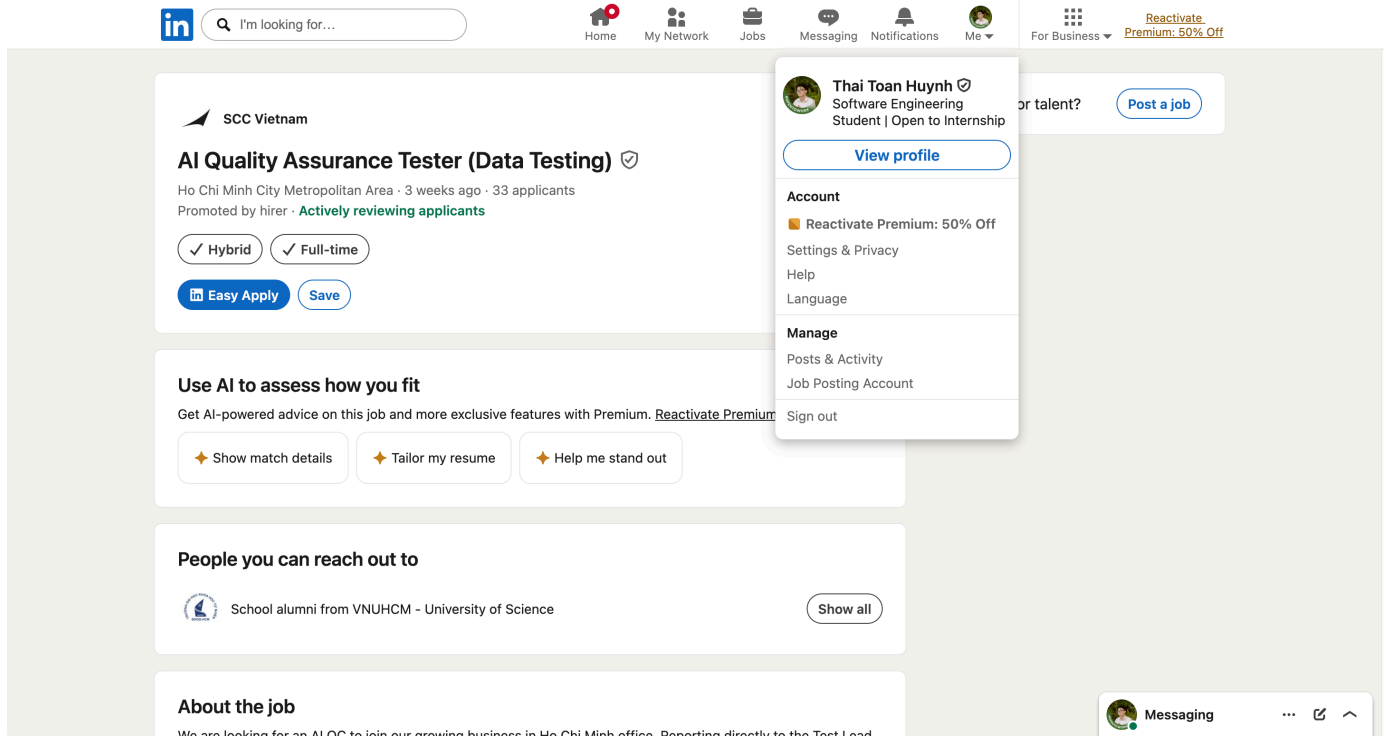
- **Link:** [ITviec job posting](#)
- **Dated screenshot:** “Posted 20 days ago”, captured 24/09/2026 (posted 04/09/2026).



- **Job description:** Ensure the quality of backend APIs in an Agile team through automated and manual testing, using AI tools to speed up test design, automation, debugging and quality analysis. Build API test scripts (Postman, Python); test an email system end to end (API, IMAP, SMTP, JMAP, migration, synchronization, authentication); run benchmarking, performance, load and stress tests; integrate API tests into Jenkins CI/CD and report results with Allure.
- **Required skills:** 4+ years of automation testing; end-to-end automation process; experience using AI tools (Claude, Cursor, Copilot, ChatGPT) to improve testing productivity; Python; API automation (Python, Java, JMeter, Postman); CI/CD (GitHub, Jenkins, GitLab CI, nightly runs); analytical and problem-solving skills. Performance and security testing are a plus.
- **Salary:** Not disclosed (net salary based on skills and experience).
- **AI Impact Analysis:** AI tools can speed up API test design, script writing and failure debugging for this role. Protocol-level behavior such as IMAP/SMTP synchronization and performance bottlenecks still needs a human tester to design realistic workloads and interpret the results.

JP02 – AI Quality Assurance Tester (Data Testing), SCC Vietnam

- **Link:** [LinkedIn job posting](#)
- **Dated screenshot:** “3 weeks ago”, captured 23/09/2026.



- **Job description:** Develop test scripts and test data for Scout, an AI-powered sales agent in Microsoft Teams. Validate chat-based AI answers, data consolidation from ZoomInfo, CRM and web sources, meeting-preparation outputs, conflict-resolution rules and EU AI Act transparency/logging; perform SIT/FAT with the UK team, report defects and support the Test Lead's documentation.
- **Required skills:** Bachelor's degree; 2–4 years in software or AI testing (manual and automated); Microsoft Teams, Azure AI/Cloud and CRM platforms (Sales Hub, Salesforce); testing conversational interfaces, data pipelines or AI-powered applications; AI conflict resolution, data governance and EU AI Act; SIT/FAT, testing lifecycle, Agile, root-cause analysis and stakeholder management.
- **Salary:** Not disclosed (competitive remuneration).
- **AI Impact Analysis:** AI can help generate data variations, classify failures and accelerate conversational test analysis. Human testers must still judge correctness, context and harmful or misleading responses.

JP03 – Quality Engineering, DXC Technology

- **Link:** [DXC Careers \(Workday\) job posting](#) – job ID 51585699
- **Dated screenshot:** “Posted 16 Days Ago”, captured 24/09/2026 (posted 08/09/2026).



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Quality Engineering

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VNM - HO CHI MINH CITY

Full time

Posted 16 Days Ago

51585699

Job Description:

DXC Technology (NYSE: DXC) is a leading enterprise technology and innovation partner delivering software, services, and solutions to global enterprises and public sector organizations — helping them harness AI to drive outcomes at a time of exponential change with speed. With deep expertise in Managed Infrastructure Services, Application Modernization, and Industry-Specific Software Solutions, DXC modernizes, secures, and operates some of the world's most complex technology estates. Learn more on [dxc.com](#)

DXC Vietnam has been recognized as one of the "Best Companies to Work for in Asia 2025" by HR Asia Magazine, highlighting its exceptional HR practices and 31-year legacy in the country. The company also earned the special "Most Caring Company Award" in 2025 for prioritizing employee well-being and fostering a supportive, inclusive workplace culture.

We are seeking motivated Senior Quality Engineers who are passionate about applying their technical skills to develop and deliver industry-leading lending solutions for small and medium-sized businesses across Globe.

About Us



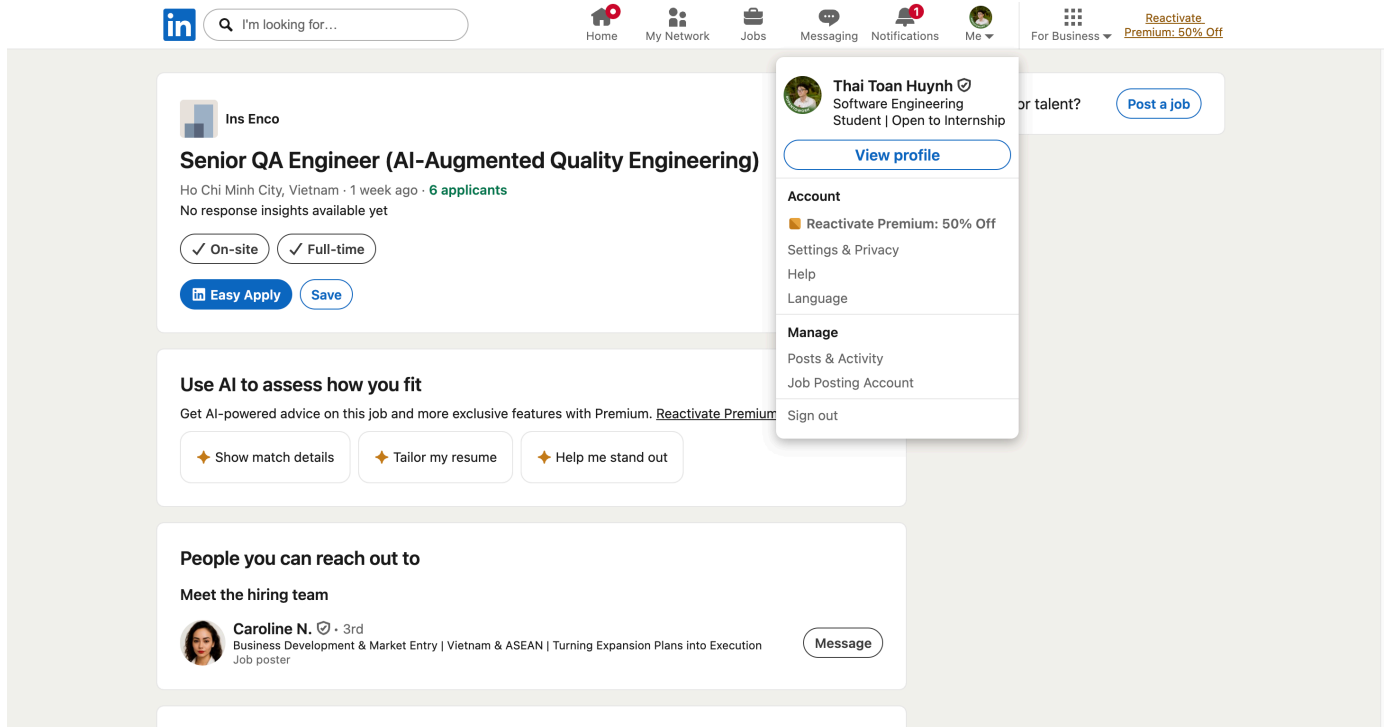
DXC Technology (NYSE: DXC) is a leading enterprise technology and innovation partner delivering software, services, and solutions to global enterprises and public sector organizations — helping them harness AI to drive outcomes at a time of exponential

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- **Job description:** Develop and run automated tests for lending software, integrate tests into CI/CD, create test plans and summary reports, review solution documents, and help set acceptance criteria and resolve defects.
- **Required skills:** 3+ years in software testing, including 2+ years of web/mobile and API automation; Selenium WebDriver with Java or Cypress with JavaScript/TypeScript; testing methods, Agile, and English communication. Postman/PACT/SoapUI/REST Assured and QTest/Jira/Confluence are also listed.
- **Salary:** Not disclosed.
- **AI Impact Analysis:** Automation tools can reduce repetitive regression work and provide earlier feedback in CI/CD. Human QA still needs to assess lending risks, acceptance criteria, test coverage and whether fixes address the underlying defects.

JP04 – Senior QA Engineer (AI-Augmented Quality Engineering), Ins Enco

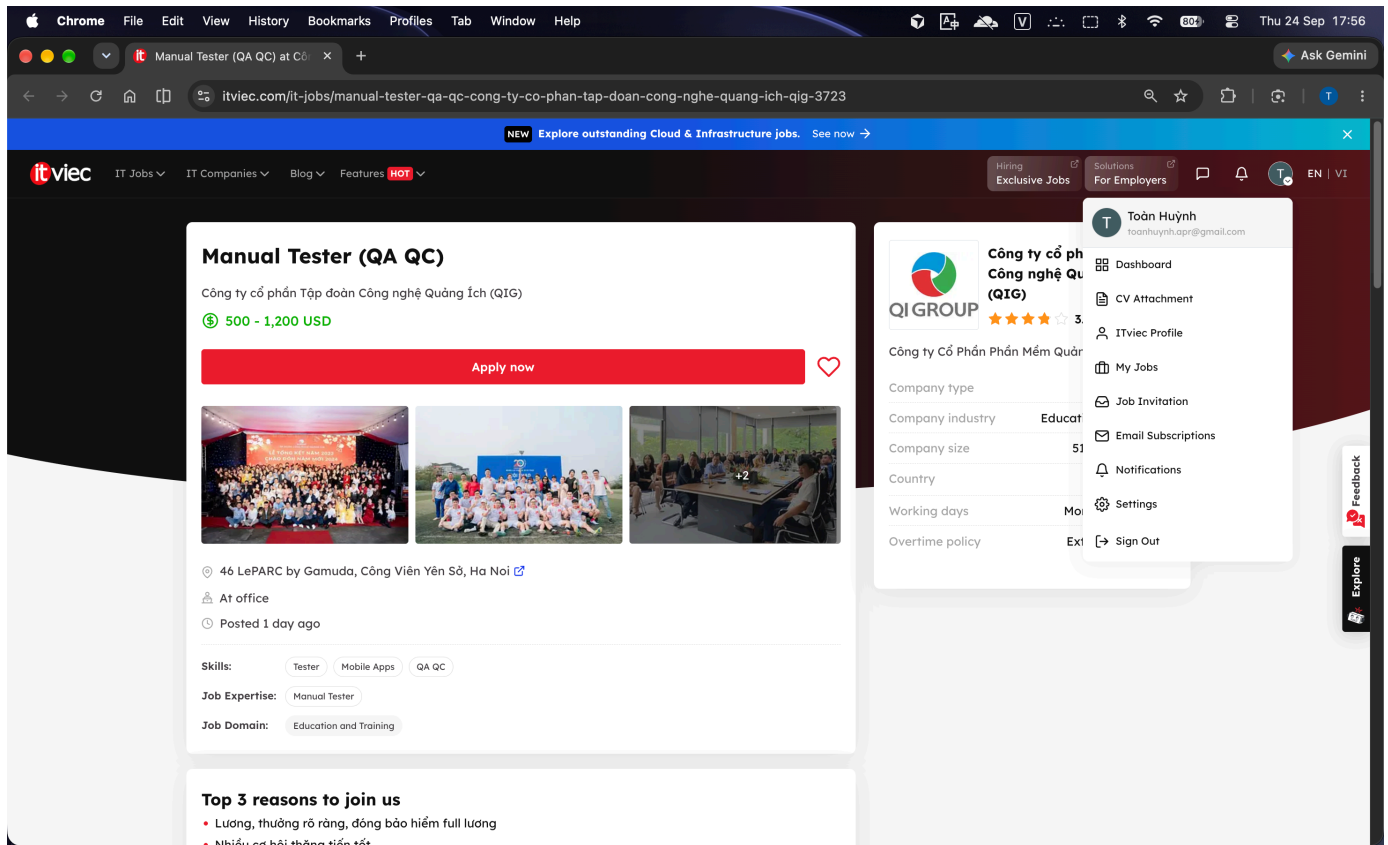
- **Link:** [LinkedIn job posting](#)
- **Dated screenshot:** “1 week ago”, captured 23/09/2026.



- **Job description:** Own quality engineering across test strategy, exploratory and risk-based testing, automation, CI/CD, production quality and AI-assisted testing. The role includes generating test cases/data with GPT or Claude, analyzing failure logs with AI and using Postbot to generate API tests.
- **Required skills:** 3–6 years of QA experience; Playwright; API testing with Postman/Bruno (REST and GraphQL); SQL/database validation; Git and CI/CD; Jira with Xray/Zephyr; Agile. AI-essential: GPT/Claude for test generation and analysis, Mabl/Testim, Applitools/Percy and Postbot. Performance, security and observability tools are nice-to-have.
- **Salary:** 30,000,000–40,000,000 VND gross/month, depending on experience.
- **AI Impact Analysis:** AI is positioned as part of test design and failure triage, so it can speed up test-data generation and log analysis. The engineer still owns risk-based coverage, review of AI suggestions and release-quality decisions.

JP05 – Manual Tester (QA QC), Công ty cổ phần Tập đoàn Công nghệ Quảng Ích (QIG)

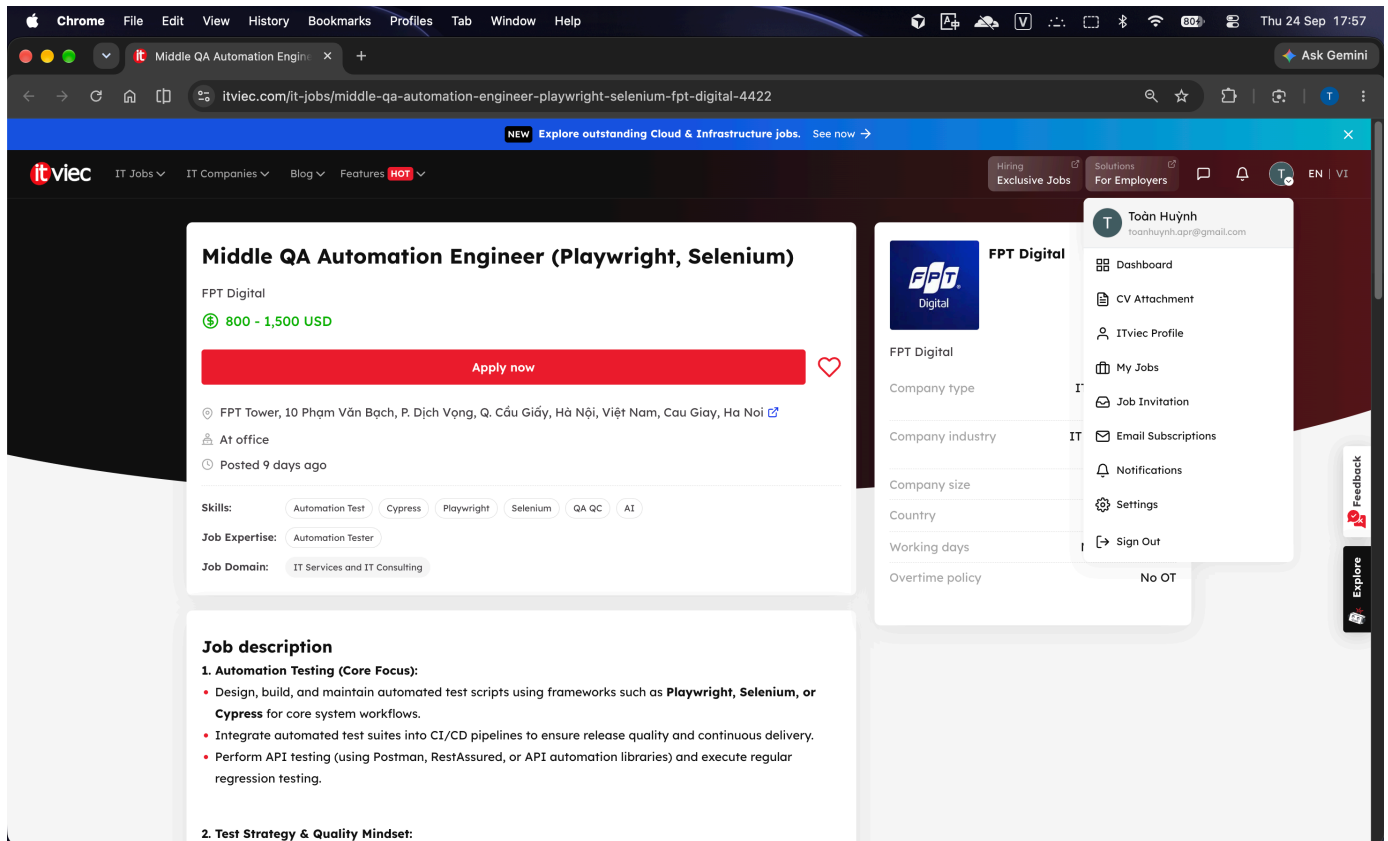
- **Link:** [ITviec job posting](#)
- **Dated screenshot:** “Posted 1 day ago”, captured 24/09/2026 (posted 23/09/2026).



- **Job description:** Write test plans and test cases from business analysis, prepare test data and environments, execute tests, log bugs with severity and priority, track and report test results, work with developers on fixes, propose test-process improvements, and hand over product knowledge to the customer-support and operations teams.
- **Required skills:** 2+ years as a tester; solid knowledge of the test process and testing techniques; teamwork. Experience testing web or mobile applications and business-heavy software is an advantage.
- **Salary:** 500–1,200 USD/month (the job description states 13–25 million VND/month, depending on experience).
- **AI Impact Analysis:** AI can assist with drafting test cases and bug reports from business requirements. Understanding complex business rules and training the support team on them still needs a human tester.

JP06 – Middle QA Automation Engineer (Playwright, Selenium), FPT Digital

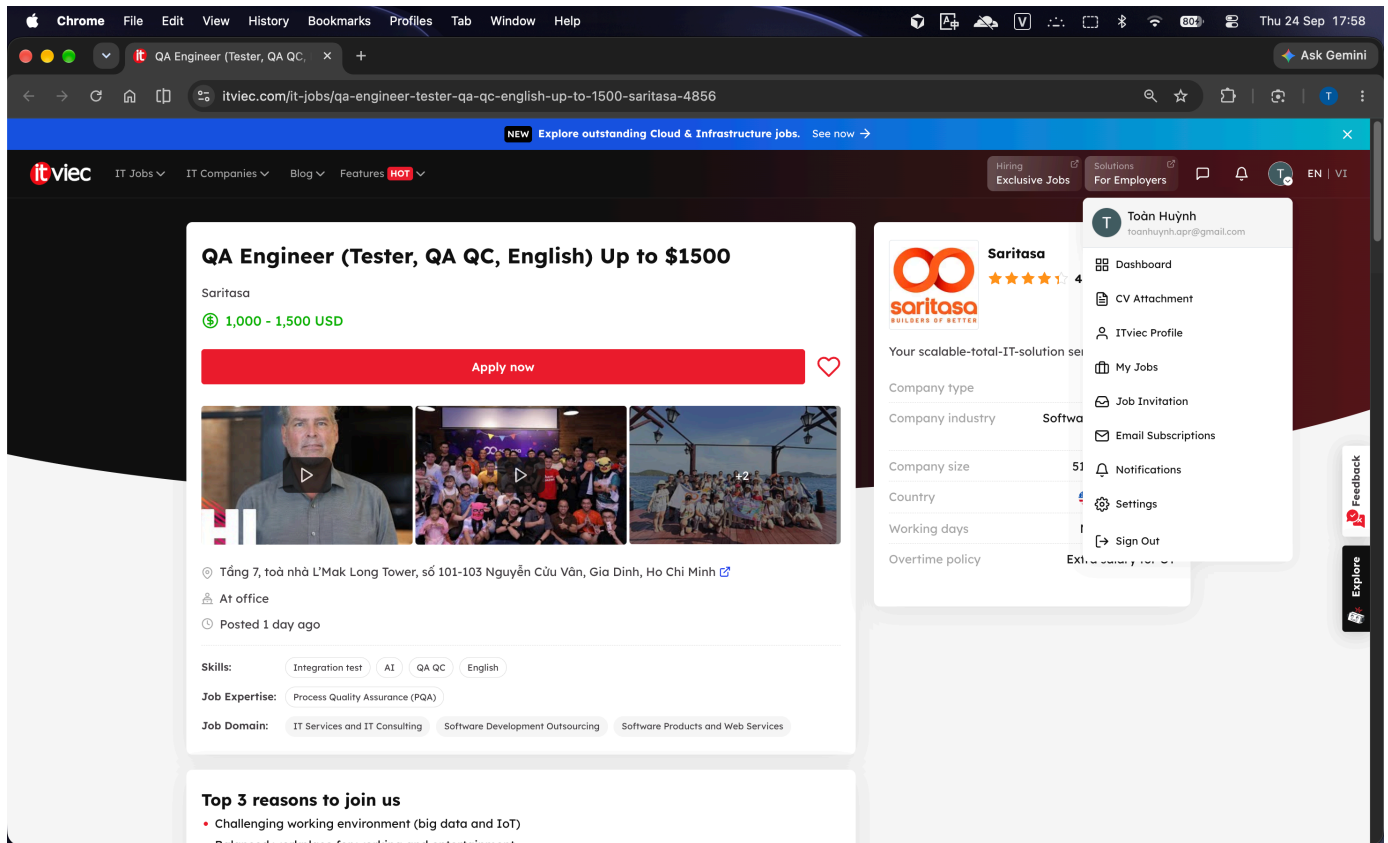
- **Link:** [ITviec job posting](#)
- **Dated screenshot:** “Posted 9 days ago”, captured 24/09/2026 (posted 15/09/2026).



- **Job description:** Build and maintain Playwright/Selenium/Cypress test scripts integrated into CI/CD; perform API, regression, functional and SQL data-integrity testing; design test cases covering happy paths, edge cases and boundary conditions; manage defects in Jira. Test AI/LLM features (chatbots, data extraction, RAG workflows) for accuracy, completeness, source attribution, hallucinations and stability across prompt variations, and use ChatGPT, Claude or Copilot to generate test cases, data and scripts.
- **Required skills:** Bachelor's in CS/IT; 2–4 years of manual and automation testing; one UI automation framework (Playwright, Selenium or Cypress) with JavaScript/TypeScript, Python or Java; API testing with Postman or automation libraries; SQL; experience testing AI/LLM features or daily use of GenAI assistants in QA work. CI/CD pipelines and RAG/prompt-engineering knowledge are pluses.
- **Salary:** 800–1,500 USD/month.
- **AI Impact Analysis:** GenAI assistants can speed up test-case, data and script generation, but they cannot be trusted to judge their own output. The tester is needed to define quality criteria for LLM features and to detect hallucinated or inconsistent answers.

JP07 – QA Engineer (Tester, QA QC, English) Up to \$1500, Saritasa

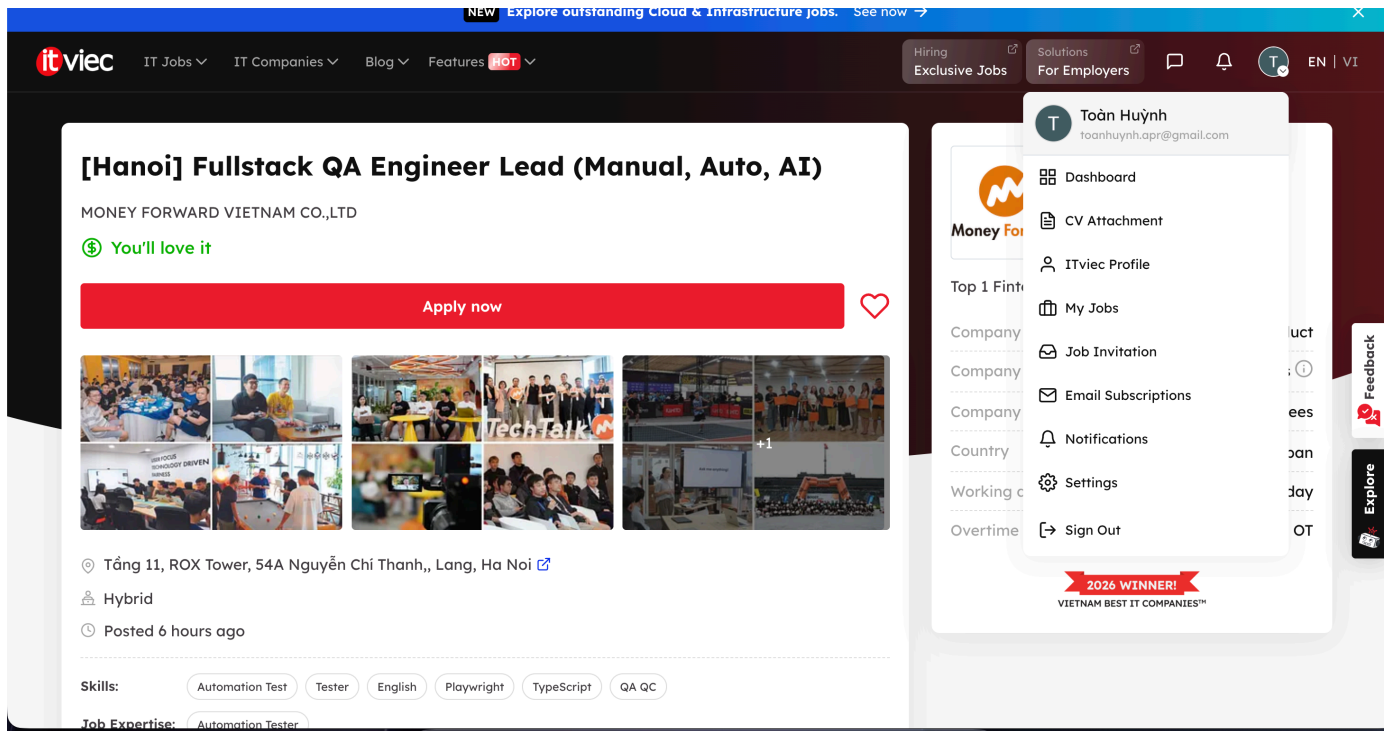
- **Link:** [ITviec job posting](#)
- **Dated screenshot:** “Posted 1 day ago” (refreshed listing), captured 24/09/2026; original posting date 16/09/2026.



- **Job description:** Test web, mobile (iOS/Android) and enterprise applications built with PHP/.NET/Python backends and React/Angular frontends for international clients, working with Russia-based teams. Analyze specifications in detail, break complex requirements into simple use cases, test APIs with Postman/JMeter, and document bugs with screenshots and videos.
- **Required skills:** 3+ years as a manual QA/tester (automation preferred); software-testing theory, principles and techniques; attention to detail and analytical thinking; good English. Preferred: web, mobile and API testing, Git, and using AI tools to speed up test design and daily work.
- **Salary:** 1,000–1,500 USD/month.
- **AI Impact Analysis:** AI tools can speed up test design and bug documentation across many client projects. Turning complex design requirements into meaningful use cases and explaining quality issues to clients in English remain human work.

JP08 – [Hanoi] Fullstack QA Engineer Lead (Manual, Auto, AI), Money Forward Vietnam

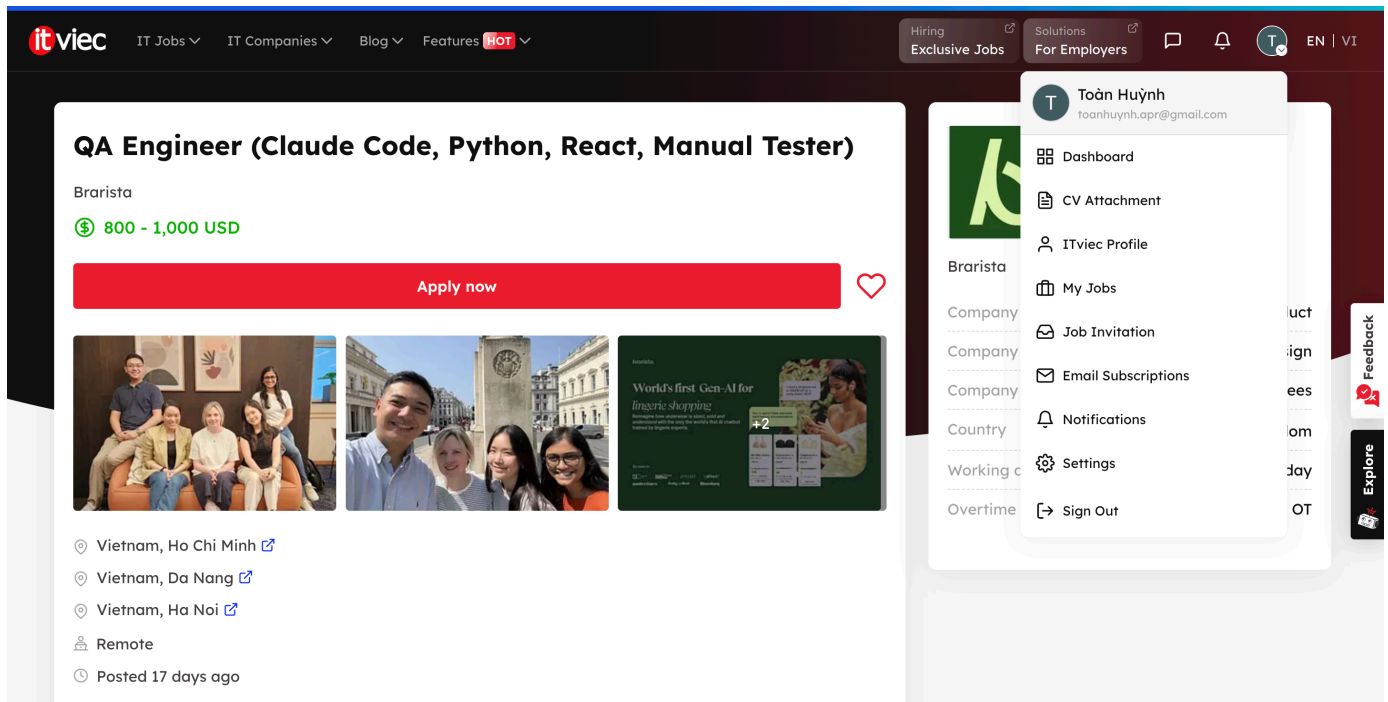
- **Link:** [ITviec job posting](#)
- **Dated screenshot:** “Posted 6 hours ago” (refreshed listing), captured 24/09/2026; original posting date 14/09/2026.



- **Job description:** Act as the quality lead for Money Forward's B2B SaaS products (Michibiku, Conkan), working with Japan-based PMs and engineers. Own release-risk decisions and entry/exit criteria, architect Playwright/TypeScript (or Python) web and API automation with CI/CD quality gates, govern outsourced QA teams, drive shift-left testing, and lead adoption of GenAI tools for test design, script generation and triage, including testing AI-assisted product features.
- **Required skills:** Bachelor's in CS/SE; 8+ years of QA/SDET experience with 3+ years in a technical leadership role; fluent English; JavaScript/TypeScript or Python with Playwright, Cypress or Selenium; REST/GraphQL API testing and SQL validation; CI/CD and Dockerized test environments; daily practical use of GenAI tooling. Japanese and performance testing are nice to have.
- **Salary:** Not disclosed.
- **AI Impact Analysis:** AI tools are expected to speed up test design, script generation and triage, and the role also tests AI-assisted features. The lead still has to decide release risk, set safe AI-usage rules and judge whether AI output is correct.

JP09 – QA Engineer (Claude Code, Python, React, Manual Tester), Brarista

- **Link:** [ITviec job posting](#)
- **Dated screenshot:** “Posted 17 days ago”, captured 24/09/2026 (posted 07/09/2026).



- **Job description:** Test Brarista's AI fit-assistance products (sizing engine, chat assistant, client dashboard and storefront widgets). Use Claude Code to generate size-conversion test matrices and regression suites, verify AI-suggested catalogue tags, own release quality and the changelog, and reproduce and triage client-reported bugs. Remote; part-time or full-time.
- **Required skills:** Strong attention to detail; proven, heavy use of Claude Code, including catching its wrong output; edge-case thinking; clear written English for bug reports and release notes. Nice to have: Intercom/Gleap, changelog writing, pytest/Playwright, e-commerce/Shopify.
- **Salary:** 800–1,000 USD/month.
- **AI Impact Analysis:** Claude Code can generate the thousands of size combinations that cannot be checked by hand. The tester must still catch wrong AI output and wrong AI-assigned tags, because these errors reach real customers without any crash or alarm.

JP10 – Software Quality Analyst (QA, Tester), MiTek Vietnam

- **Link:** [ITviec job posting](#)
- **Dated screenshot:** “Posted 33 days ago”, captured 23/09/2026 (posted 21/08/2026).

Software Quality Analyst (QA, Tester)

MiTek Vietnam

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🏢 At office

🕒 Posted 33 days ago

Skills: QA QC Automation Test Tester JavaScript Python English

Job Expertise: Automation Tester

Job Domain: Real Estate Property and Construction Software Products and Web Services

User Profile: Toàn Huỳnh (toanhuynh.apr@gmail.com)

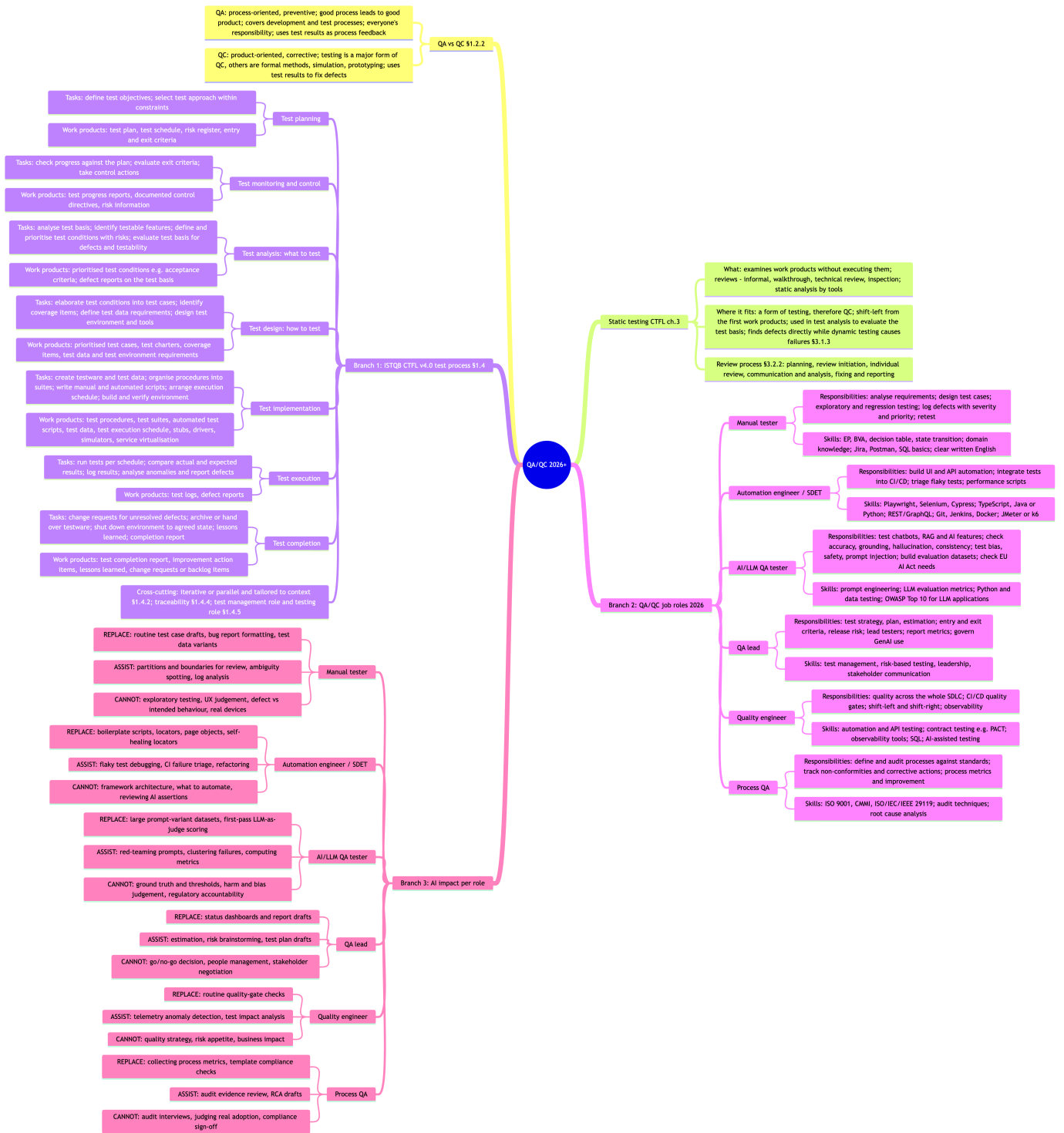
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- Notifications
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- Sign Out

- **Job description:** Design, maintain and execute functional, regression and automation test cases; develop automated test scripts for Web, Windows applications and APIs; track defects through the testing lifecycle; debug and refactor tests; work with teams in Vietnam and the US to clarify requirements and designs.
- **Required skills:** 4+ years of manual and automated testing; test automation for Web, Windows applications and APIs with any framework and language; Agile; testing processes and techniques; English communication; source control. SQL, Postman, CI/CD, Python and home-building domain knowledge are pluses.
- **Salary:** Not disclosed (competitive salary).
- **AI Impact Analysis:** AI can assist with repetitive quality analysis and report drafting, while human analysts remain responsible for interpreting business impact and communicating quality decisions.

1.4 QA/QC mindmap and its mistakes (G9.1)

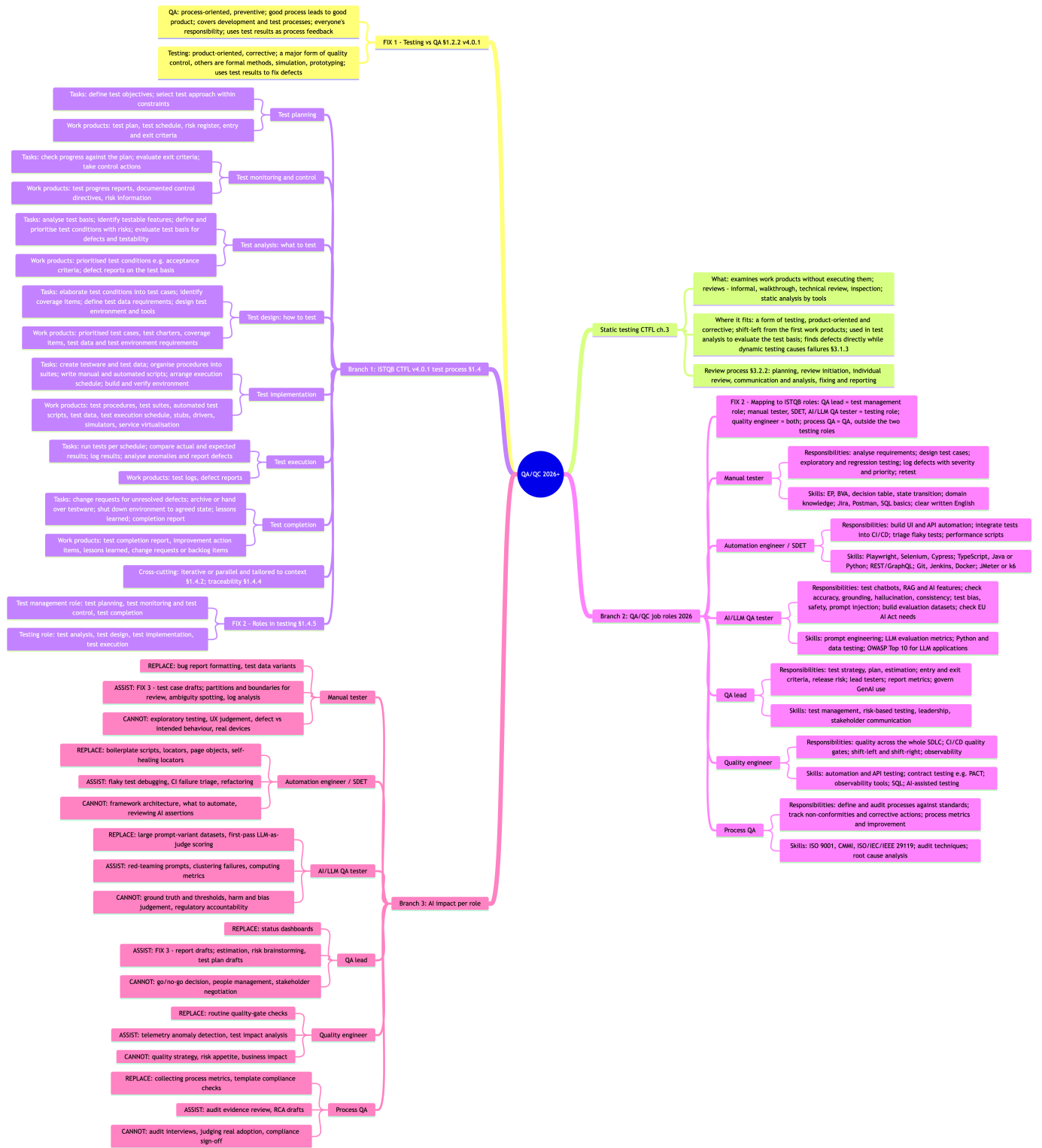
The mindmap was generated by Cursor Agent (Claude Opus 5.5) at 18:08 24/09/2026 (prompt log entry R1-M01); the raw output is in R1_Mindmap/Mindmap_AI_Output.md. It was checked against the current ISTQB CTFL syllabus v4.0.1 (15/09/2024) and the earlier v4.0 (21/04/2023). The test activities, testware, review process, review types and the placement of exit-criteria evaluation (§5.3) are correct. Three mistakes were found.

AI-generated mindmap (original):



#	AI claim (quoted)	What the syllabus says	Type
1	Node "QA vs QC §1.2.2": "QC: product-oriented, corrective; testing is a major form of QC ... uses test results to fix defects"; static testing is "a form of testing, therefore QC".	This is the wording of the old v4.0. In the current v4.0.1, §1.2.2 is titled "Testing and Quality Assurance (QA)" and says: "Testing is a product-oriented, corrective approach ... Testing is a major form of quality control" and "Test results are used by QA and testing. In testing they are used to fix defects". A mindmap for "QA/QC 2026+" cites an outdated syllabus version.	Outdated source
2	Only "test management role and testing role §1.4.5" as a cross-cutting note; Branch 2 lists six job titles with no link to these roles.	§1.4.5: the test management role "is mainly focused on the activities of test planning, test monitoring, test control and test completion"; the testing role "is mainly focused on the activities of test analysis, test design, test implementation and test execution". The mindmap never maps activities to roles or job titles to roles, although the prompt asked to connect the test process with QA/QC roles.	Incomplete
3	Branch 3: "Manual tester – REPLACE: routine test case drafts" and "QA lead – REPLACE: status dashboards and report drafts". The legend defines REPLACE as output that needs "only spot-check review by a human".	The evidence from this homework contradicts it. ChatGPT's 15 test cases (R3-T02) had 10 incomplete and 1 invalid case and missed 5 edge cases (section 3.3). Its defect explanations (R2) had confirmed errors in 10 of 20 defects (section 2.4). Test design must derive coverage items with test techniques (§1.4.1, chapter 4), and test reports inform stakeholder decisions (§5.3.2). Both need full human review, which is ASSIST, not REPLACE.	Wrong classification (over-trust in AI)

Corrected mindmap (changes labelled "FIX 1", "FIX 2", "FIX 3"; source R1_Mindmap/QAQC_2026_Mindmap_Corrected.mmd):



- **FIX 1:** node renamed to "Testing vs QA §1.2.2 v4.0.1" with the v4.0.1 definitions; Branch 1 now cites v4.0.1.
- **FIX 2:** new node "Roles in testing §1.4.5" mapping the activities to the two ISTQB roles, and a Branch 2 node mapping each job title to these roles.
- **FIX 3:** "test case drafts" and "report drafts" moved from REPLACE to ASSIST.

Requirement 2 – Software Defects 2022–2026 (20 points)

2.1 Scope and severity scale

- **Number of defects:** 20, all publicized between 2022 and 2025.
- **AI/LLM-related defects:** 7 (D14–D20): hallucination (D14, D15, D17), bias (D16), prompt injection (D18, D19, D20). D13 occurred in an LLM service but was caused by a conventional library bug, so it is not counted as AI/LLM-related.
- **Sources:** vendor advisories and post-incident reports, the U.S. National Vulnerability Database (NVD), regulators (FCC, CRTC, UK CAA, NHTSA, FAA) and court or tribunal decisions.
- **Severity:** for CVEs, the CVSS v3.1 base score from NVD (and the vendor score when it differs). For incidents without a CVSS score, severity is assessed with the scale below.

Severity	Criteria used for incidents
Critical	Nationwide or global outage of essential services (aviation, emergency calls, telecommunications), or risk to life safety
High	Large-scale outage or data exposure, safety risk mitigated by human supervision, or harmful output delivered to a mass audience
Medium	Financial or legal harm to individual users; limited scope
Low	Cosmetic or easily worked-around issue

2.2 Summary of defects

ID	Defect	Product / system	Publicized	AI/LLM	Severity	Primary source
D01	Faulty Channel File 291 content update crashes Windows hosts	CrowdStrike Falcon sensor	19/07/2024	No	Critical	CrowdStrike PIR
D02	BGP policy reordering withdraws prefixes in 19 data centers	Cloudflare network	21/06/2022	No	High	Cloudflare blog
D03	Routing filter removed during core network upgrade	Rogers Communications	08/07/2022	No	Critical	CRTC report
D04	Damaged database file takes down NOTAM system	FAA NOTAM system	11/01/2023	No	Critical	FAA statement
D05	Duplicate waypoint names crash flight-plan processing	NATS FPRSA-R	28/08/2023	No	Critical	CAA independent review
D06	Misconfigured network element blocks all wireless traffic	AT&T Mobility network	22/02/2024	No	Critical	FCC report
D07	FSD Beta violates traffic laws at intersections	Tesla Full Self-Driving Beta	16/02/2023	No	High	NHTSA recall 23V-085
D08	SQL injection in MOVEit Transfer web app (CVE-2023-34362)	Progress MOVEit Transfer	31/05/2023	No	Critical (9.8)	NVD
D09	Citrix Bleed buffer over-read leaks session tokens (CVE-2023-4966)	NetScaler ADC / Gateway	10/10/2023	No	Critical (vendor 9.4) / High (NVD 7.5)	NVD
D10	X.509 email address 4-byte buffer overflow (CVE-2022-3602)	OpenSSL 3.0.0–3.0.6	01/11/2022	No	High (7.5)	OpenSSL advisory
D11	Backdoor in xz/liblzma release tarballs (CVE-2024-3094)	XZ Utils 5.6.0–5.6.1	29/03/2024	No	Critical (10.0)	NVD
D12	GlobalProtect command injection (CVE-2024-3400)	Palo Alto PAN-OS	12/04/2024	No	Critical (10.0)	Palo Alto advisory
D13	redis-py bug shows other users' chat titles and payment data	OpenAI ChatGPT	24/03/2023	LLM service, non-model bug	High	OpenAI post-mortem
D14	ChatGPT fabricates legal cases cited in a court filing	ChatGPT (Mata v. Avianca)	22/06/2023	Yes – hallucination	High	Court opinion

ID	Defect	Product / system	Publicized	AI/LLM	Severity	Primary source
D15	Airline chatbot invents a bereavement refund policy	Air Canada website chatbot	14/02/2024	Yes – hallucination	Medium	BC CRT decision 2024 BCCRT 149
D16	Image generator overcorrects for diversity, producing inaccurate historical images	Google Gemini	23/02/2024	Yes – bias	High	Google blog
D17	AI Overviews present satire and forum jokes as facts	Google Search AI Overviews	30/05/2024	Yes – hallucination	High	Google blog
D18	Prompt injection leads to code execution in LLMChain (CVE-2023-29374)	LangChain ≤ 0.0.131	05/04/2023	Yes – prompt injection	Critical (9.8)	NVD
D19	EchoLeak zero-click prompt injection exfiltrates Copilot data (CVE-2025-32711)	Microsoft 365 Copilot	11/06/2025	Yes – prompt injection	Critical (vendor 9.3) / High (NVD 7.5)	NVD
D20	Prompt injection in ask visualization leads to RCE (CVE-2024-5565)	Vanna.AI	31/05/2024	Yes – prompt injection	High (8.1)	JFrog advisory

2.3 Detailed defect records

D01 – CrowdStrike Falcon Channel File 291

- **Source:** [CrowdStrike Preliminary Post Incident Review](#); [Microsoft blog](#), 20/07/2024
- **Description:** On 19/07/2024 a Rapid Response Content update (Channel File 291) was pushed to Falcon sensors on Windows. A bug in the Content Validator let a template instance with problematic content pass validation. When the sensor loaded it, the content caused an out-of-bounds memory read in the CSagent.sys kernel driver, crashing Windows (blue screen) and preventing normal boot.
- **Severity:** Critical – kernel-level crash of production hosts worldwide, including airlines, hospitals and banks.
- **Consequences:** Microsoft estimated 8.5 million Windows devices were affected. Many machines needed manual recovery (booting into Safe Mode and deleting the channel file).
- **Solution:** CrowdStrike reverted the channel file about 78 minutes after release. Announced fixes: additional Content Validator checks, local developer and fault-injection testing of content, staggered (canary) deployment of Rapid Response Content, customer control over update timing, and improved error handling in the Content Interpreter.

D02 – Cloudflare outage of 21/06/2022

- **Source:** [Cloudflare outage on June 21, 2022](#)
- **Description:** A change to BGP prefix-advertisement policies re-ordered policy terms on the spine routers of the 19 data centers using the new Multi-Colo PoP (MCP) architecture. The site-local terms moved behind an explicit REJECT-THE-REST term, so critical site-local prefixes were withdrawn. The change had been dry-run and peer reviewed, but the staged rollout steps were too large to catch the error.
- **Severity:** High – these locations were only 4% of the network but handled 50% of total requests.
- **Consequences:** From 06:27 to 07:42 UTC, users in affected regions could not reach websites and services behind Cloudflare. Servers could not reach customer origins, and the internal load balancer stopped working.
- **Solution:** Engineers reverted the change router by router. Cloudflare committed to including MCP data centers earlier in its stagger policy with MCP-specific test and deploy procedures, redesigning the policy statement to prevent incorrect ordering, and automating commit-confirm rollback.

D03 – Rogers Communications nationwide outage

- **Source:** [CRTC, Assessment of Rogers Networks for Resiliency and Reliability Following the 8 July 2022 Outage](#)
- **Description:** During the sixth phase of a core IP network upgrade, a configuration change removed a routing filter. Without it, the distribution routers flooded the core routers with routes beyond their capacity, and the core network stopped processing traffic.
- **Severity:** Critical – nationwide loss of wireless and wireline services, including 911 access.
- **Consequences:** More than 12 million customers lost wireless and wireline services from 04:58 EDT on 08/07/2022 until services were gradually restored by 07:00 EDT on 09/07/2022. Calls to 911 were disrupted.
- **Solution:** The CRTC report recommends implementing router overload protection in the IP core and distribution networks, lab-testing planned configuration changes on equipment that reflects production, a more effective cross-team audit of configuration changes, limiting the number of changes per maintenance window with automatic rollback, and physically and logically separating the network management layer from the data network.

D04 – FAA NOTAM system outage

- **Source:** [FAA NOTAM statement](#)
- **Description:** The Notice to Air Missions (NOTAM) system failed because of a damaged database file. The FAA's preliminary review found that contract personnel unintentionally deleted files while working to correct synchronization between the live primary database and the backup database.
- **Severity:** Critical – safety-related information system for all U.S. flights became unavailable.
- **Consequences:** The FAA ordered airlines to pause all domestic departures until 09:00 Eastern Time on 11/01/2023 to validate the integrity of flight and safety information, causing widespread delays and cancellations.
- **Solution:** The FAA repaired the system and stated that it had taken steps to make the NOTAM system more resilient. From a testing perspective, the root cause points to the need for controlled, verified procedures when maintaining the primary and backup databases.

D05 – NATS flight-plan processing failure (FPRSA-R)

- **Source:** [CAA, Independent Review of NATS \(En Route\) Plc's Flight Planning System Failure on 28 August 2023 – Final Report](#); [NATS Major Incident Investigation Final Report](#)
- **Description:** FPRSA-R received a validly formatted flight plan for a flight from Los Angeles to Paris Orly with a unique combination of attributes involving two identical waypoint names outside UK airspace. The software constructed an invalid UK route, raised a critical exception and entered maintenance mode. The backup system then processed the same flight plan and failed in the same way.
- **Severity:** Critical – UK air traffic control lost automatic flight-plan processing nationwide.
- **Consequences:** Flight plans had to be entered manually, traffic was heavily restricted, and the CAA estimated that over 700,000 passengers were affected by cancellations and delays.
- **Solution:** NATS immediately added engineering procedures and message filtering. The permanent fix is a software change by the manufacturer that stops FPRSA-R from searching back beyond the entry waypoint into UK airspace, preventing the critical exception from recurring.

D06 – AT&T wireless network outage

- **Source:** [FCC Public Safety and Homeland Security Bureau report](#)

- **Description:** During a network expansion, a network element was misconfigured and the change was not peer reviewed as AT&T's procedures required. The configuration error caused AT&T's network to enter "protect mode", disconnecting all devices from the wireless network.
- **Severity:** Critical – nationwide loss of voice, data and 911 service.
- **Consequences:** More than 125 million registered devices were affected, more than 92 million voice calls were blocked and more than 25,000 calls to 911 could not be completed. Rolling back the change took close to two hours, and full restoration took at least 12 hours because device registration systems were overwhelmed.
- **Solution:** AT&T rolled back the network change. After the outage it added peer-review steps and procedures ensuring maintenance work cannot proceed without confirmation that the required peer reviews were completed. The FCC also cited inadequate lab testing and post-installation testing as contributing factors.

D07 – Tesla FSD Beta recall 23V-085

- **Source:** [NHTSA Part 573 Safety Recall Report 23V-085](#)
- **Description:** With FSD Beta engaged, the vehicle could infringe local traffic laws in four situations: travelling or turning through certain intersections during a stale yellow light; not stopping long enough at stop signs; adjusting speed incorrectly in variable speed zones; and changing lanes out of certain turn-only lanes to continue straight.
- **Severity:** High – increased collision risk if the driver does not intervene; the risk is limited by the SAE Level 2 requirement that the driver constantly supervises.
- **Consequences:** 362,758 vehicles (2016–2023 Model S and X, 2017–2023 Model 3, 2020–2023 Model Y) were recalled. Tesla identified 18 possibly related warranty claims and no known injuries or deaths.
- **Solution:** Free over-the-air software update that improves how FSD Beta handles these maneuvers.

D08 – MOVEit Transfer SQL injection (CVE-2023-34362)

- **Source:** [NVD CVE-2023-34362](#); [CISA/FBI advisory AA23-158A](#)
- **Description:** An SQL injection vulnerability in the MOVEit Transfer web application allowed an unauthenticated attacker to access the database, infer its structure, and execute SQL statements that alter or delete data.
- **Severity:** Critical – CVSS 9.8; exploited as a zero-day and listed in CISA's Known Exploited Vulnerabilities catalog.
- **Consequences:** From 27/05/2023 the CL0P ransomware gang exploited the flaw to install the LEMURLOOT web shell on internet-facing servers and steal data from MOVEit databases of many organizations.
- **Solution:** Upgrade to a fixed release (2021.0.6, 2021.1.4, 2022.0.4, 2022.1.5, 2023.0.1 or later). Progress also advised disabling HTTP/HTTPS traffic to MOVEit until patched and hunting for indicators of compromise such as human2.aspx.

D09 – Citrix Bleed (CVE-2023-4966)

- **Source:** [NVD CVE-2023-4966](#); [Citrix bulletin CTX579459](#)
- **Description:** A buffer over-read in NetScaler ADC and Gateway configured as a gateway (VPN, ICA proxy, CVPN, RDP proxy) or AAA virtual server disclosed memory contents, including valid session tokens.
- **Severity:** Critical according to the vendor (9.4); High according to NVD (7.5).
- **Consequences:** Attackers hijacked authenticated sessions and bypassed multi-factor authentication. The flaw was exploited in the wild, including by LockBit 3.0 ransomware affiliates.

- **Solution:** Upgrade to 14.1-8.50, 13.1-49.15, 13.0-92.19 or later (FIPS/NDcPP builds as listed in the bulletin), then kill all active and persistent sessions, because stolen tokens remain valid after patching.

D10 – OpenSSL X.509 buffer overflow (CVE-2022-3602)

- **Source:** [OpenSSL Security Advisory](#), [01/11/2022](#)
- **Description:** During X.509 certificate name-constraint checking, a crafted email address could overflow four attacker-controlled bytes on the stack. It could be triggered by a TLS client connecting to a malicious server, or by a server requesting client authentication.
- **Severity:** High (CVSS 7.5). It was pre-announced as Critical and downgraded to High after analysis of mitigating factors such as stack protections.
- **Consequences:** Possible crash (denial of service) or, in theory, remote code execution. OpenSSL reported no known working exploit at release time.
- **Solution:** Upgrade OpenSSL 3.0.x to 3.0.7. OpenSSL 1.1.1 and 1.0.2 were not affected.

D11 – XZ Utils backdoor (CVE-2024-3094)

- **Source:** [NVD CVE-2024-3094](#); [CISA alert](#)
- **Description:** Malicious code was hidden in the upstream release tarballs of xz 5.6.0 and 5.6.1. Obfuscated build steps extracted a prebuilt object file from test files and modified liblzma, which can then interfere with software that links it, such as sshd.
- **Severity:** Critical – CVSS 10.0; a deliberate supply-chain backdoor.
- **Consequences:** Affected packages reached development and rolling distributions (Fedora 41/Rawhide, Debian testing/unstable, openSUSE Tumbleweed, Kali, Arch). It was detected before it reached most stable enterprise releases.
- **Solution:** Downgrade to an uncompromised version such as XZ Utils 5.4.6 (CISA), install distribution-fixed packages, and hunt for malicious activity on systems that had the affected versions.

D12 – PAN-OS GlobalProtect command injection (CVE-2024-3400)

- **Source:** [Palo Alto Networks advisory CVE-2024-3400](#)
- **Description:** Arbitrary file creation in the GlobalProtect feature of PAN-OS 10.2, 11.0 and 11.1 led to OS command injection, allowing an unauthenticated attacker to execute code with root privileges on the firewall.
- **Severity:** Critical – CVSS 10.0; discovered in production use.
- **Consequences:** An increasing number of attacks exploited the flaw, and proof-of-concept exploits were published.
- **Solution:** Upgrade to PAN-OS 10.2.9-h1, 11.0.4-h1, 11.1.2-h3 or the listed hotfixes. As interim mitigation, apply Threat Prevention signatures (Threat IDs 95187, 95189, 95191) to the GlobalProtect interface. Compromised devices can use an enhanced factory reset.

D13 – ChatGPT data exposure on 20/03/2023

- **Source:** [OpenAI, March 20 ChatGPT outage: Here's what happened](#)
- **Description:** A bug in the open-source redis-py client library caused cancelled requests to leave corrupted connections. The next request on such a connection could receive cached data belonging to a different user.
- **Severity:** High – exposure of personal and partial payment data.
- **Consequences:** Some users saw other users' chat history titles. For about 1.2% of ChatGPT Plus subscribers active during a nine-hour window, name, email, payment address, the last four digits of the card number and

the card expiration date could have been visible to another user.

- **Solution:** OpenAI took ChatGPT offline, patched the bug with the Redis maintainers, added redundant checks that returned data belongs to the requesting user, and improved logging.

D14 – ChatGPT hallucinated case law (Mata v. Avianca)

- **Source:** [Mata v. Avianca, Inc., Opinion and Order on Sanctions, S.D.N.Y., 22/06/2023](#)
- **Description:** Lawyers used ChatGPT for legal research and submitted a brief citing six judicial decisions that did not exist. ChatGPT generated fake case names, quotations and internal citations, and when asked, stated that the cases were real.
- **Severity:** High – fabricated authority submitted to a federal court.
- **Consequences:** The court sanctioned the attorneys and their firm with a USD 5,000 penalty and required them to send letters to their client and to each judge falsely named as an author of the fake opinions.
- **Solution:** The court held that lawyers remain responsible for verifying every citation. The corrective practice is to check AI-generated citations against authoritative legal databases before filing.

D15 – Air Canada chatbot misrepresentation (Moffatt v. Air Canada)

- **Source:** [Moffatt v. Air Canada, 2024 BCCRT 149](#)
- **Description:** Air Canada's website chatbot told a customer that, after travelling, they could submit their ticket for a reduced bereavement rate within 90 days of the ticket issue date. The airline's bereavement policy page said the policy does not apply after travel has been completed, and an Air Canada representative later admitted the chatbot had used "misleading words".
- **Severity:** Medium – financial harm to an individual customer, but it set a legal precedent.
- **Consequences:** The tribunal found Air Canada liable for negligent misrepresentation, rejected the argument that the chatbot was responsible for its own actions, and ordered the airline to pay CAD 812.02 (damages, interest and fees).
- **Solution:** The company is accountable for all information on its website, including chatbot answers. The chatbot must be grounded in and tested against the official policy pages.

D16 – Gemini image generation bias

- **Source:** [Google, Gemini image generation got it wrong. We'll do better \(23/02/2024\)](#)
- **Description:** Tuning intended to show a range of people did not account for prompts that clearly should not show a range, such as specific historical contexts. Over time the model also became over-cautious and refused some benign prompts.
- **Severity:** High – inaccurate and biased output from a mass-market AI product.
- **Consequences:** Gemini generated historically inaccurate images of people, causing public criticism and reputational damage to Google.
- **Solution:** Google paused image generation of people in Gemini and committed to extensive testing before re-enabling it.

D17 – Google AI Overviews misleading answers

- **Source:** [Google, AI Overviews: About last week \(30/05/2024\)](#)
- **Description:** AI Overviews generated odd and wrong answers for nonsensical queries and "data voids", sometimes relying on satirical content or forum posts as if they were factual.
- **Severity:** High – misleading information shown at the top of Google Search to a mass audience.

- **Consequences:** Widely shared examples of absurd advice damaged trust in the feature and in Google Search.
- **Solution:** Google made more than a dozen technical improvements, including better detection of nonsensical queries, limiting satire and humor content, limiting user-generated content that could give misleading advice, adding triggering restrictions, and strengthening guardrails for news and health topics.

D18 – LangChain LLMMathChain prompt injection (CVE-2023-29374)

- **Source:** [NVD CVE-2023-29374](#)
- **Description:** In LangChain through 0.0.131, LLMMathChain passed LLM-generated text to Python exec. A prompt injection in the user's question could make the LLM output arbitrary Python code, which was then executed.
- **Severity:** Critical – CVSS 9.8.
- **Consequences:** Any application exposing LLMMathChain to untrusted input allowed remote code execution on its server.
- **Solution:** Upgrade to LangChain 0.0.142 or later, where LLMMathChain no longer uses exec and evaluates expressions with numexpr (later constrained to numexpr $\geq$ 2.8.6, which sanitizes input).

D19 – EchoLeak in Microsoft 365 Copilot (CVE-2025-32711)

- **Source:** [NVD CVE-2025-32711](#); [EchoLeak case study \(arXiv 2509.10540\)](#).
- **Description:** AI command injection in M365 Copilot allowed an unauthorized attacker to disclose information over a network. A crafted email containing hidden instructions was retrieved as context by Copilot, bypassed the cross-prompt-injection classifier and link redaction, and caused Copilot to embed sensitive data in an auto-fetched image URL, with no user interaction.
- **Severity:** Critical according to Microsoft (9.3); High according to NVD (7.5).
- **Consequences:** Potential zero-click exfiltration of any data in Copilot's context. Microsoft reported no evidence of exploitation in the wild and no affected customers.
- **Solution:** Microsoft deployed a server-side fix in May 2025, before public disclosure; no customer action was required.

D20 – Vanna.AI prompt injection to RCE (CVE-2024-5565)

- **Source:** [JFrog advisory JFSA-2024-001034449](#); [NVD CVE-2024-5565](#)
- **Description:** Vanna's ask method, with visualize=True (the default), asks the LLM to produce Plotly code and then executes it. A prompt injection in the question can make the LLM return arbitrary Python code instead of chart code.
- **Severity:** High – CVSS 8.1.
- **Consequences:** Remote code execution on any server that passes external user input to ask.
- **Solution:** When ask receives external input, set visualize=False. More generally, never execute LLM-generated code outside a sandbox.

2.4 AI explanation errors (bias or hallucination)

Method: ChatGPT (GPT-5.6 Luna) was asked to explain all 20 defects in two batch prompts (R2-B01, R2-B02), followed by two rounds of more specific follow-up questions (R2-B03, R2-B04) for defects whose answers contained no error. The 10 defects that still had no error were asked again, as one batch, of Cursor Composer (composer-2.5-fast) on 27/09/2026 (R2-B05). Every claim was compared with the sources in section 2.3 (URLs were also checked for HTTP status).

Result: confirmed errors were found for all 20 defects. Ten came from ChatGPT. The other ten came from Composer.

ID	Prompt log entry (HH:MM dd/mm/yyyy)	AI claim (quoted)	Verified fact and source	Error type
D01	R2-B03, 23:29 24/09/2026	Cites “CrowdStrike RCA” with the URL of falcon-content-update-preliminary-post-incident-report .	That URL is the Preliminary Post Incident Review (24/07/2024); the RCA was published separately on 06/08/2024 (CrowdStrike).	Hallucination (misattributed source)
D02	R2-B05, 22:23 27/09/2026	“widespread HTTP 500 errors”	The Cloudflare post does not mention HTTP 500. It says users could not reach sites, site-local prefixes were withdrawn, and the internal load balancer stopped (Cloudflare).	Hallucination (invented symptom)
D03	R2-B05, 22:23 27/09/2026	Source: about.rogers.com/news-ideas/rogers-2022-service-interruption-report/ .	The URL returns HTTP 404. The outage assessment is the CRTC report .	Hallucination (fabricated source)
D04	R2-B01, 23:19 24/09/2026	“The FAA issued a nationwide ground stop around 07:30 EST and lifted it around 09:00.”	The FAA's 7:15 a.m. EST statement already reported the pause of domestic departures, and its 8:50 a.m. EST statement said the ground stop had been lifted (FAA).	Hallucination (timeline)
D05	R2-B01, 23:19 24/09/2026	FPRSA-R is the “Flight Planning and Resiliency System for Airspace”.	FPRSA-R is the “Flight Plan Reception Suite Automated” sub-system (NATS final report).	Hallucination (invented acronym expansion)
D06	R2-B01, 23:19 24/09/2026	“AT&T restored service by around noon.”	The outage began at 2:45 AM CST; AT&T announced restoration at 2:10 PM CST, and full restoration took over 12 hours (FCC).	Hallucination (timeline)
D07	R2-B05, 22:23 27/09/2026	The recall covered “2017–2023 Model 3/Y”.	NHTSA 23V-085 lists Model Y as 2020–2023, not 2017–2023 (NHTSA).	Hallucination (wrong population)
D08	R2-B05, 22:23 27/09/2026	Fixed releases “2020.1.11, 2021.0.7, 2022.0.7, 2022.1.8, and 2023.0.2”.	Progress fixed CVE-2023-34362 in 2021.0.6, 2021.1.4, 2022.0.4, 2022.1.5 and 2023.0.1 (Progress).	Hallucination (wrong version)

ID	Prompt log entry (HH:MM dd/mm/yyyy)	AI claim (quoted)	Verified fact and source	Error type
D09	R2-B01, 23:19 24/09/2026	The flaw “is commonly rated CVSS 3.1: 9.4 (Critical)”; the official fix was to upgrade to patched builds.	9.4 is only the vendor score; NVD rates it 7.5 High (NVD). NetScaler also required killing all active and persistent sessions after upgrading (NetScaler).	Bias (one-sided score) and incomplete fix
D10	R2-B05, 22:23 27/09/2026	The fix versions are “3.0.7, 1.1.1s, and 1.0.2zg”.	The advisory says only 3.0.0–3.0.6 are affected, 1.1.1 and 1.0.2 are not, and the fix is 3.0.7 (OpenSSL).	Hallucination (wrong version)
D11	R2-B05, 22:23 27/09/2026	Source: cisa.gov/.../recommended-actions-mitigate-xz-utils-supply-chain-compromise .	That URL returns HTTP 404. CISA’s alert says downgrade to 5.4.6, not reissue 5.6.2 (CISA).	Hallucination (fabricated source)
D12	R2-B03, 23:29 24/09/2026	Source: https://www-dev.volexity.com/blog/2024/04/12/...	The host www-dev.volexity.com does not resolve in DNS (checked 24/09/2026); the article exists on www.volexity.com .	Hallucination (fabricated source)
D13	R2-B05, 22:23 27/09/2026	“1.2% of Plus users had title exposure”.	OpenAI says 1.2% is the share whose payment-related information may have been visible. Chat titles were a separate exposure and were not given that percentage (OpenAI).	Hallucination (wrong statistic)
D14	R2-B05, 22:23 27/09/2026	Source: nysd.uscourts.gov/sites/default/files/23cv1461_20230622_order.pdf .	The URL returns HTTP 404. The opinion is on Justia .	Hallucination (fabricated source)
D15	R2-B02, 23:19 24/09/2026	The chatbot “told a passenger that he could obtain a bereavement-fare refund”; “awarding C\$812.02 in damages and fees”.	The decision refers to Jake Moffatt only as “they”; C\$812.02 = C\$650.88 damages + C\$36.14 interest + C\$125 fees (2024 BCCRT 149 , paras. 42–44).	Bias (gender assumption) and inaccurate breakdown
D16	R2-B02, 23:19 24/09/2026	Source: blog.google/products/gemini/gemini-image-generation-people-update/ .	The URL returns HTTP 404; Google’s explanation is at gemini-image-generation-issue (23/02/2024).	Hallucination (fabricated source)

ID	Prompt log entry (HH:MM dd/mm/yyyy)	AI claim (quoted)	Verified fact and source	Error type
D17	R2-B02, 23:19 24/09/2026	“On 31 May 2024, Google said it had made more than a dozen technical improvements”, citing .../generative-ai-google-search-may-2024/.	Google's post “AI Overviews: About last week” is dated 30 May 2024 (Google); the cited URL is the I/O launch announcement.	Hallucination (date and source)
D18	R2-B03, 23:29 24/09/2026	The vulnerability “was fixed in 0.0.132”.	The fix is in LangChain 0.0.142 (Snyk); NVD lists affected versions up to 0.0.131 only.	Hallucination (wrong version)
D19	R2-B05, 22:23 27/09/2026	Customers should “apply Microsoft’s prescribed configuration mitigations plus documented patch”.	Microsoft deployed a server-side fix and required no customer action (NVD ; report section 2.3).	Hallucination (wrong fix)
D20	R2-B05, 22:23 27/09/2026	“NVD assigned CVSS v3.1 9.8 (Critical)” and the fix is “upgrading to patched Vanna releases”.	NVD has not scored this CVE. JFrog scored it 8.1 High. The documented mitigation is visualize=False; NVD lists no patched release (NVD).	Hallucination (invented score and fix)

Requirement 3 – Physical Product Testing (25 points)

3.1 Device under test

Field	Value
Device	Standing electric fan
Brand	Senko
Model	B813
Year of manufacture	2014
Serial number (middle 4 characters masked)	RA****6H
Speed control	Rotary knob with marked positions 0 (Off), 1, 2, 3
Oscillation control	Push-pull knob on the motor housing
Other adjustments	None
Test environment	Bedroom, level tiled floor, 220 V household outlet, room temperature ~30 °C, tested 27/09/2026



3.2 Test approach

- **State transition testing:** the fan has four speed states (Off, 1, 2, 3) and two oscillation states (on, off). Because the speed control is a rotary knob, moving between non-adjacent positions ($1 \rightarrow 3$, $3 \rightarrow 1$, $3 \rightarrow 0$) passes through the intermediate positions; test cases cover both rotation directions.
- **Decision table:** speed $\times$ oscillation combinations (TC-09), and power state $\times$ knob position (TC-12).

- **Boundary value analysis:** the knob's end positions 0 and 3 (TC-07), the lowest speed started from standstill (TC-02), and the strongest load, Speed 3 with oscillation, for stability and endurance (TC-13, TC-15).
- **Error guessing / edge cases:** knob resting between two detents, turning beyond the end positions, oscillation changed while the fan is off, mechanical resistance at the head, and guard safety (TC-06, TC-07, TC-10, TC-11, TC-14).
- **Measurable expected results:** where no manufacturer specification is available, the tester-defined thresholds are stated in the test case so the verdict is not subjective.
- **Safety limits:** no disassembly, no contact with moving blades, no electrical fault injection. Guard checks are done unplugged.

3.3 Evaluation of the initial AI output (R3-T02)

ChatGPT produced 15 test cases from the corrected prompt (prompt log entry R3-T02). An earlier attempt (R3-T01) used an ambiguous control description and is kept in the prompt log only. Each R3-T02 test case was reviewed against the device and against black-box testing practice.

AI TC	Verdict	Reasoning
TC01	VALID	Correct, observable check of the Off position.
TC02	INCOMPLETE	"Low-speed airflow" is not measurable; it does not check that the blades start unaided from standstill at the lowest speed, the hardest start condition for an aged motor.
TC03	INCOMPLETE	"Noticeably stronger than level 1" gives no measurement method.
TC04	INCOMPLETE	Same problem as TC03.
TC05	VALID	The sequence $1 \rightarrow 2 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0$ matches how a rotary knob is operated.
TC06	INVALID	The objective is a "direct transition" from 0 to 2 and 0 to 3, which a sequential rotary knob cannot perform: the knob always passes through the lower positions. The case also duplicates TC02–TC04.
TC07	INCOMPLETE	Assumes "knob UP = oscillation off" without a source, and duplicates TC09.
TC08	INCOMPLETE	Assumes "push DOWN = on" without a source; no sweep angle or period is checked.
TC09	INCOMPLETE	Same assumption as TC08; "remains in a fixed position" gives no stop criterion (how soon the head must stop).
TC10	VALID	Speed $\times$ oscillation combination is covered.
TC11	INCOMPLETE	Pass criteria ("excessive force", "ambiguous operation") are subjective; label legibility is not checked. The precondition "Fan OFF" conflicts with the step of rotating through 1–3, which runs the fan.
TC12	INCOMPLETE	"An extended period, e.g. 30–60 min" is unbounded; "sparking" cannot be observed from outside the closed switch; it runs at Speed 2, not at the maximum load.
TC13	INCOMPLETE	"Excessive shaking" and "tipping tendency" are undefined; it is listed under reliability although it is a stability (safety) check.
TC14	INCOMPLETE	Visual inspection only, while the blades are spinning; it never checks whether a finger can actually reach a blade, which can be checked safely with the fan unplugged.
TC15	VALID	Observable shutdown and restart.

Summary: VALID 4/15 (26.7%), INVALID 1/15 (6.6%), INCOMPLETE 10/15 (66.7%); total 100%.

Other issues in the output:

- It twice claims that "public references" and "public documentation for the B813 family" confirm the design, but only shows citation markers (: chatgpt-content-reference) with no document that can be checked. These claims are unverifiable and were not used.
- It assumes the oscillation direction (push = on) without asking.
- It has no case for power interruption, although a mechanical knob keeps its position when power is lost.

3.4 Edge cases missed by the AI

None of the following cases appear in R3-T02.

Final TC	Edge case	Why the AI missed it
TC-06	Speed knob left between two detents	The prompt listed four discrete positions, so ChatGPT modelled the knob as four valid states and only tested those. A physical knob can also rest between marks; this invalid input partition exists on the device but not in the text description, so the AI never derived it.
TC-07	Speed knob turned beyond its end positions 0 and 3	ChatGPT treated "0–3" as the complete input domain and tested values inside it, but not outside the boundaries. Whether the knob stops at 3 or wraps to 0 is not in the prompt, and the AI did not ask; boundary value analysis requires testing just beyond each end.
TC-10	Oscillation knob changed while the fan is off	Every oscillation case in R3-T02 has the precondition "fan running". ChatGPT tested the two controls in one order only (speed first, then oscillation) and did not enumerate the combination speed 0 + oscillation on, which a user can easily set.
TC-11	Head held or turned by hand while oscillating	ChatGPT deliberately kept safety cases "observation-based" and avoided physical interaction with the fan. It also only considered inputs through the two controls, not external forces on the head, such as a child or a curtain stopping it, which the oscillation gear must tolerate.
TC-12	Power interruption with the knob left at a speed position	ChatGPT considered only the user controls as inputs, not the environment (mains power). Because a mechanical knob keeps its position, the fan restarts on its own when power returns, for example during the night or while someone is cleaning it. This risk only appears when the tester thinks about the device's state after an outage.

Evidence: screenshots of the R3-T02 conversation — R3_Device/ai_output_R3-T02.png.

3.5 Final test cases

Verdict values: PASS, FAIL, NOT RUN. Test cases that have not been executed are marked NOT RUN.

ID	Objective	Preconditions	Input	Steps	Expected result	Actual result	Verdict	Technique	Edge (AI-missed)	Evidence
TC-01	Controls are identifiable and labels legible	Fan unplugged	Visual inspection; turn the speed knob through every position	1. Without the manual, identify the Off, 1, 2, 3 marks and the oscillation knob. 2. Read each mark and the knob pointer from 1 m. 3. Turn the speed knob through every position.	Every position mark and the pointer are readable; each position has a distinct detent (click) so the user can feel where it stops.	NOT RUN	NOT RUN	Usability	No	–
TC-02	Lowest speed starts from standstill	Fan plugged in, blades fully stopped for ≥ 5 min, knob at 0	Knob to 1	1. Turn the knob to 1. 2. Start stopwatch. 3. Stop it when blades reach steady rotation.	Blades start unaided (no hand push) and reach steady rotation within 5 s. Start time recorded.	NOT RUN	NOT RUN	BVA	No	–
TC-03	Speeds are distinct and increasing	Fan running, oscillation off, paper strip hung 1 m in front of the guard	Knob at 1, 2, 3	1. Run 30 s at each speed. 2. Record the strip deflection and the sound level (phone app, 1 m). 3. Repeat once.	Deflection and sound level increase strictly from 1 to 2 to 3.	NOT RUN	NOT RUN	EP	No	–
TC-04	Speed transitions in both rotation directions	Fan running at 1	1→2→3, then 3→2→1; 1→3 and 3→1 in one continuous turn	1. Turn step by step up, then down, pausing 10 s at each position. 2. Turn 1→3 and 3→1 in one continuous movement. 3. Observe the speed after each stop.	The speed always matches the position the knob stops at, with no stop, hum or stall during the turn.	27/09/2026. Turning the knob in one movement from 1 to 3, the motor clicks one beat each time the pointer passes the midpoint between two marks. When the knob stops on a mark, the speed matches that mark.	FAIL	State transition	No	–
TC-05	Off from each speed and coast-down	Fan running	Knob to 0 from 1, 2, 3	1. At each speed turn the knob to 0. 2. Time until the blades stop.	The motor is de-energised as soon as the knob reaches 0; the blades coast to a stop; times recorded and longer for higher speeds.	NOT RUN	NOT RUN	State transition	No	–
TC-06	Knob left between two detents	Fan at 0	Knob stopped halfway between 0–1, 1–2, 2–3	1. Turn the knob slowly and stop midway between two marks. 2. Release it and observe the knob and the motor for 10 s.	The knob snaps to one of the two marked positions, or the fan runs steadily at one of the two speeds; no intermittent start/stop, hum or stall.	27/09/2026. When the knob is left between two speed marks, the fan switches off.	FAIL	Error guessing	Yes	HW01-TC06

ID	Objective	Preconditions	Input	Steps	Expected result	Actual result	Verdict	Technique	Edge (AI-missed)	Evidence
TC-07	Knob turned beyond its end positions	Fan at 3	Gentle extra turn past 3; then from 0, gentle turn backwards past 0	1. At 3, keep turning gently in the same direction. 2. At 0, turn gently in the opposite direction. 3. Observe the knob and the fan.	The knob behaves as marked on the panel: it either stops firmly at the end mark or moves to the next marked position. It never rests at an unmarked position, and the fan speed always matches the marked position.	27/09/2026. Turning past 3 moves the knob to 0 and the fan switches off. Turning onward from 0 moves the knob to the next marked position. The speed matches the mark.	PASS	BVA	Yes	HW01-TC07
TC-08	Oscillation on/off, sweep angle and period	Fan at 2, oscillation off, tape marks on the floor for the head direction	Oscillation knob	1. Toggle oscillation on. 2. Time three full sweeps. 3. Mark both end positions and measure the angle. 4. Toggle off.	Head sweeps smoothly and symmetrically; period consistent across the three sweeps (± 1 s); stops within one sweep after toggling off. Angle and period recorded.	27/09/2026. Switching oscillation off while the head is sweeping, the head sweeps more than one extra turn before it stops.	FAIL	State transition	No	–
TC-09	Oscillation at each speed	Oscillation on	1, 2, 3	1. Run one full sweep at each speed. 2. Listen for clicking or grinding.	Oscillation continues at every speed with no sticking, stopping or grinding noise.	27/09/2026. The head keeps sweeping at speeds 1, 2 and 3, with no sticking or grinding.	PASS	Decision table	No	HW01-TC09
TC-10	Oscillation set while the fan is off	Fan at 0, oscillation off	Oscillation knob on, then speed knob to 1	1. With the fan off, set oscillation on. 2. Confirm the motor does not start. 3. Turn the speed knob to 1.	Changing the oscillation knob does not start the motor; after turning to 1, the fan runs and oscillates immediately.	27/09/2026. Setting oscillation while the fan is off does not start it. After turning to speed 1, the fan runs and oscillates.	PASS	State transition	Yes	HW01-TC10
TC-11	Head held or turned by hand	Fan at 1, hand on the rear motor housing only (never the guard)	Gentle hold for 2 s during oscillation; gentle turn with oscillation off	1. With oscillation on, hold the head still for 2 s, then release. 2. With oscillation off, turn the head 20° and release.	Step 1: the gear slips without grinding and oscillation resumes after release. Step 2: the head stays at the new position.	27/09/2026. Holding the head still for 2 seconds while it is oscillating, the gears grind. After release the head stays still and oscillation does not resume.	FAIL	Error guessing	Yes	–
TC-12	Behaviour after power interruption	Fan at 2	Unplug 10 s, plug in; repeat with knob at 0	1. With the knob at 2, unplug and replug. 2. Turn the knob to 0, unplug and replug.	Step 1: the fan restarts at Speed 2 as soon as power returns (the knob stays at 2). Step 2: the fan stays off.	27/09/2026. With the knob at 2, the fan restarts at speed 2 when power returns. With the knob at 0, it stays off.	PASS	Decision table	Yes	HW01-TC12
TC-13	Stability at maximum load	Level floor, tape around the base	3 + oscillation on, 5 min	1. Run for 5 min. 2. Measure any base displacement against the tape.	No tipping or rocking; the base moves less than 1 cm.	NOT RUN	NOT RUN	BVA	No	–

ID	Objective	Preconditions	Input	Steps	Expected result	Actual result	Verdict	Technique	Edge (AI-missed)	Evidence
TC-14	Finger cannot reach the blades	Fan unplugged, blades stopped	Ruler; finger test at the front and rear guard	1. Measure the largest gap in the front and rear guard. 2. Try to touch a blade with a fingertip through the gap. 3. Pull the guard clips gently by hand.	A fingertip cannot reach a blade; the guard stays closed when pulled by hand.	27/09/2026. With the fan unplugged, a fingertip cannot touch a blade through the front guard. Through the rear guard a fingertip can touch a blade.	FAIL	Error guessing	No (AI TC14 is visual only)	–
TC-15	Endurance for 60 min	Clear area, room temperature recorded	3 + oscillation on, 60 min	1. Run 60 min. 2. At 0, 30 and 60 min record speed, oscillation, noise, and whether the motor housing and cord can be touched comfortably.	Continuous operation and oscillation; no burning smell or smoke; housing and cord warm but comfortable to touch; noise level stable.	NOT RUN	NOT RUN	Reliability	No	–

3.6 Execution evidence and defects

TC	Video (YouTube Unlisted, ≤ 60 s, narrated)
TC-06	HW01-TC06
TC-07	HW01-TC07
TC-09	HW01-TC09
TC-10	HW01-TC10
TC-12	HW01-TC12

Defect ID	TC	Summary	Expected	Actual	Severity	GitHub Issue
DF-01	TC-06	The fan switches off when the speed knob rests between two marks.	The knob snaps to one marked position, or the fan runs steadily at one of the two speeds.	When the knob is left between two marks, the fan switches off. Observed 27/09/2026 on the Senko B813.	Medium	#1
DF-02	TC-04	The motor clicks one beat each time the speed knob passes a midpoint while turning from 1 to 3.	The speed matches the position the knob stops at, with no stop, hum or stall during the turn.	Each time the pointer passes the midpoint between two marks, the motor clicks one beat. When the knob stops on a mark, the speed matches that mark. Observed 27/09/2026 on the Senko B813.	Medium	#2
DF-03	TC-08	The head sweeps more than one extra turn after oscillation is switched off.	The head stops within one sweep after oscillation is switched off.	Switching oscillation off while the head is sweeping, the head sweeps more than one extra turn before it stops. Observed 27/09/2026 on the Senko B813.	Medium	#3
DF-04	TC-11	The oscillation gear grinds when the head is held, and the head stays still after release.	The gear slips without grinding, and oscillation resumes after release.	Holding the head still for 2 seconds while it is oscillating, the gears grind. After release the head stays still and oscillation does not resume. Observed 27/09/2026 on the Senko B813.	High	#4
DF-05	TC-14	A fingertip can touch a blade through the rear guard.	A fingertip cannot reach a blade.	With the fan unplugged, a fingertip cannot touch a blade through the front guard. Through the rear guard a fingertip can touch a blade. Observed 27/09/2026 on the Senko B813.	High	#5

Screenshot of the GitHub Issues page. The page shows username Toanapr and issues #1–#5.

The screenshot shows the GitHub interface for the repository 'Toanapr / Homework'. The 'Issues' tab is selected, showing 5 open issues. The issues are listed in descending order of creation time, all opened by 'Toanapr' 2m ago, except for the first one which was opened 1h ago. The issues are:

- DF-05 Senko B813 rear guard lets a fingertip touch a blade
- DF-04 Senko B813 oscillation gear grinds and the head stays still after a 2-second hold
- DF-03 Senko B813 head sweeps more than one extra turn after oscillation is switched off
- DF-02 Senko B813 motor clicks one beat each time the speed knob passes a midpoint from 1 to 3
- DF-01 Senko B813 fan switches off when the speed knob rests between two marks

The left sidebar contains navigation icons for repository overview, issues, pull requests, actions, projects, and security. The top navigation bar includes links to code, issues, pull requests, actions, projects, and security, along with a search bar and a 'Public' label.

Issue #1, opened while logged in as Toanapr:

Toanapr / Homework

[Code](#)
[Issues](#)
[Pull requests](#)
[Agents](#)
[Actions](#)
[Projects](#)
[Wiki](#)
[Security and quality](#)
[Insights](#)
[Settings](#)

DF-01 Senko B813 fan switches off when the speed knob rests between two marks #1

New issue

Open

Toanapr opened 3m ago

Owner

Test case: TC-06
Device: Senko B813 standing fan, manufactured 2014, serial RA****6H
Date: 27/09/2026
Environment: bedroom, level tiled floor, 220 V household outlet

Steps:

1. Start with the speed knob at 0.
2. Turn the knob slowly and leave it between two speed marks.
3. Release the knob and watch for about 10 seconds.

Expected: The knob snaps to one marked position, or the fan runs steadily at one of the two speeds.
Actual: When the knob is left between two marks, the fan switches off.
Severity: Medium
Evidence: https://youtube.com/shorts/k_u6U1VcS8w

Create sub-issue

Assignees

No one - [Assign yourself](#)

Assign to Agent

Labels

No labels

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

[Create a branch](#) for this issue or link a pull request.

Add a comment

3.7 Test summary

Copied from the "Test Summary" sheet of HW01_Test_Workbook.xlsx (sheets: Test Summary, Test Cases, Checklist, Defects). Status on 27/09/2026, after TC-04, TC-06, TC-07, TC-08, TC-09, TC-10, TC-11, TC-12 and TC-14.

Metric	Value	Target
Test cases designed	15	15
Executed (PASS + FAIL)	9	≥ 5
PASS / FAIL / BLOCKED	4 / 5 / 0	–
NOT RUN	6	–
Videos recorded	5	≥ 5
Edge cases missed by the AI	5	≥ 3
Defects logged	5	aim ≥ 5

Technique	Designed	Executed
State transition	4	3
BVA	3	1
Error guessing	3	3
Decision table	2	2
EP	1	0
Usability	1	0
Reliability	1	0

Five defects are open: DF-01 and DF-02 (speed knob between marks), DF-03 (oscillation stops late), DF-04 (oscillation gear grinds and does not resume) and DF-05 (rear guard). DF-04 and DF-05 are High because one can damage the gear and the other lets a finger reach a blade. TC-04, TC-08, TC-11 and TC-14 have no video yet; the defect records use the observation from 27/09/2026.

AI Critique

Across this homework the AI was most useful as a fast first draft and least reliable exactly where a tester has to be precise. In Requirement 2, ChatGPT (GPT-5.6 Luna) explained all 20 defects fluently, and 10 explanations contained a confirmed error: shifted timelines (FAA, AT&T), an invented expansion of the acronym FPRSA-R, a wrong fix version for LangChain, source URLs that returned 404 or did not resolve, and a one-sided CVSS score for Citrix Bleed. A weaker model was then wrong on each of the other 10. The pattern is that the AI reproduces the shape of a correct answer and fills gaps with plausible details instead of admitting uncertainty. In Requirement 3, ChatGPT produced 15 neat test cases, but most expected results could not be measured ("noticeably stronger", "excessive shaking"), one case required a direct 0 → 3 jump that a rotary knob cannot make, and when my first prompt was ambiguous it silently assumed piano keys instead of asking. The coding assistant I used, Cursor, made the same assumption in its first draft until I corrected it. Both missed edge cases that only appear when you handle the real device: a knob resting between detents, turning past the end positions, setting oscillation while the fan is off, holding the head, and a power cut with the knob left on a speed. They missed them because they reason from the text description and from typical test suites, not from the physical input space, and ChatGPT deliberately avoided physical interaction for safety. The principle I learned is to treat AI output as an untrusted draft: verify every fact against a primary source, demand measurable expected results, and derive edge cases from the device itself.

Mandatory Disclosure

"The QA/QC mindmap, the draft report text for Requirements 1 and 2, the initial and refined test cases for Requirement 3, and the drafts of the AI Audit Report and AI Critique were initially generated by ChatGPT (GPT-5.6 Luna), Cursor Agent (Claude Opus 5.5) and Codex (OpenAI); I reviewed and modified sections 1.3–1.4 (including the corrected mindmap), 2.3–2.4 and 3.3–3.5, added edge cases TC-06, TC-07, TC-10, TC-11 and TC-12 (proposed with Cursor and checked by me against the real Senko B813 fan); the Actual result and Verdict columns of section 3.5 were written from my own observations. On 27/09/2026 I reported the failures of TC-04, TC-08, TC-11 and TC-14; Cursor formatted those observations into sections 3.5–3.7, the workbook and GitHub Issues DF-02 to DF-05. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category below."

Prohibited category (not AI-generated): the device photo with my student ID card, the execution videos with my own voice narration, the 10 job-posting screenshots showing my login name, and the prompt log timestamps.

Appendix A – AI Audit Report and Prompt Log

- **A1:** AI Audit Report, below; the same content is in the filled AI-02 template, [AI-02] – FIT@HCMUS – AI Audit Report_En – HW01.docx.
- **A2:** full prompt log with timestamps, prompt_log.md.
- The signed AI-03 Disclosure Form and AI-05 Checklist are [AI-03] – FIT@HCMUS – AI Disclosure Form_En – HW01.docx and [AI-05] – FIT@HCMUS – AI Privacy Checklist_En – HW01.docx.

A1.1 Student information

Field	Value
Student name (printed)	Huỳnh Thái Toàn
Student ID	23120175
Class / Cohort	CQ2023/31
Assignment ID	HW01 (HW01AI)
Assignment date	26/09/2026
AI tool(s) used	ChatGPT (GPT-5.6 Luna); Cursor Agent (Claude Opus 5.5); Codex (OpenAI)
AI used	☒ Yes ☐ No

A1.2 Audit table – one entry per artifact

One batch generated by a single prompt counts as one artifact. Every prompt is recorded in Appendix A2 (prompt_log.md). ISTQB references are to the CTFL v4.0 syllabus.

Artifact #1 – R1 QA/QC mindmap

1. **Prompt + tool:** Cursor Agent (Claude Opus 5.5), 18:08 24/09/2026, entry R1-M01.
2. **AI output:** R1_Mindmap/QAQC_2026_Mindmap.png and .mmd.
3. **Verdict:** INCOMPLETE
4. **Reasoning:** The test activities, testware and review process match ISTQB CTFL v4.0.1 §1.4.1, §1.4.3 and §3.2.2, but three mistakes were found (section 1.4): the QA/QC definitions follow the outdated v4.0 wording of §1.2.2; the activities and job titles are not mapped to the two roles of §1.4.5; and test case drafts and report drafts are classified as REPLACE although the R2 and R3 evidence shows they need full human review (§1.4.1, §5.3.2).
5. **Student fix:** Corrected mindmap R1_Mindmap/QAQC_2026_Mindmap_Corrected.png, with the changes labelled FIX 1–3.

Artifact #2 – R1 job-posting records (revisions)

1. **Prompt + tool:** Cursor Agent (Claude Opus 5.5), 16:59–17:59 24/09/2026. The prompts are recorded in Appendix A2 (prompt_log.md).
2. **AI output:** report sections 1.2–1.3 (commits 7356154–b113c6e).
3. **Verdict:** INCOMPLETE
4. **Reasoning:** The draft contained fields the specification does not ask for, placeholder descriptions, postings without valid posting-date evidence, and two expired postings (JP08, JP09). These were found by reviewing the work product against the assignment specification, as in the review process of ISTQB §3.2.

5. **Student fix:** Removed the extra fields; replaced JP01 and JP05–JP09 with postings inside the 60-day window; recaptured dated screenshots showing the login name.

Artifact #3 – R2 defect records D01–D20

1. **Prompt + tool:** Cursor Agent (Claude Opus 5.5), 18:14 24/09/2026. The prompt is recorded in Appendix A2 (prompt_log.md).
2. **AI output:** report sections 2.1–2.3 (commit a3bb017).
3. **Verdict:** VALID
4. **Reasoning:** Each record links to a primary source (vendor report, NVD, regulator or court decision). All 20 defects fall in 2022–2025, 7 are AI/LLM-related (≥ 5 required), and each has the description, severity and impact attributes of a defect report (ISTQB §5.5). On 27/09/2026 the sources in section 2.3 were spot-checked against the primary pages used in section 2.4. The records were kept.
5. **Student fix:** None.

Artifact #4 – R2-B01, AI explanations of D01–D10

1. **Prompt + tool:** ChatGPT (GPT-5.6 Luna), 23:19 24/09/2026, entry R2-B01.
2. **AI output:** verbatim in entry R2-B01.
3. **Verdict:** INCOMPLETE
4. **Reasoning:** 4 of 10 explanations contain verified errors: the FAA ground-stop timeline (D04), an invented expansion of FPRSA-R (D05), the AT&T restoration time (D06), and a one-sided CVSS score with an incomplete fix (D09). A defect report must be correct and traceable to its source (ISTQB §5.5), so every claim was checked against the primary source.
5. **Student fix:** Corrected facts in section 2.3; errors recorded in section 2.4.

Artifact #5 – R2-B02, AI explanations of D11–D20

1. **Prompt + tool:** ChatGPT (GPT-5.6 Luna), 23:19 24/09/2026, entry R2-B02.
2. **AI output:** verbatim in entry R2-B02.
3. **Verdict:** INCOMPLETE
4. **Reasoning:** 3 of 10 explanations contain verified errors: a gender assumption and an amount given without its breakdown (D15), and fabricated or mismatched source URLs and dates (D16, D17). Same basis as artifact #5 (ISTQB §5.5).
5. **Student fix:** Corrected facts and sources in section 2.3; errors recorded in section 2.4.

Artifact #6 – R2-B03, follow-up questions

1. **Prompt + tool:** ChatGPT (GPT-5.6 Luna), 23:29 24/09/2026, entry R2-B03.
2. **AI output:** verbatim in entry R2-B03.
3. **Verdict:** INCOMPLETE
4. **Reasoning:** 3 of 13 answers contain verified errors: a misattributed CrowdStrike source (D01), a non-resolving `www-dev.volexity.com` host (D12), and a wrong LangChain fix version, 0.0.132 instead of 0.0.142 (D18).
5. **Student fix:** Correct sources and version recorded in section 2.4.

Artifact #7 – R2-B04, second follow-up

1. **Prompt + tool:** ChatGPT (GPT-5.6 Luna), 23:43 24/09/2026, entry R2-B04.

2. **AI output:** verbatim in entry R2-B04.
3. **Verdict:** VALID
4. **Reasoning:** Every checked claim matched the primary sources. The Scribd source cited for D14 could not be checked, so it is not counted as an error.
5. **Student fix:** None.

Artifact #8 – R3-T01, first 15 test cases

1. **Prompt + tool:** ChatGPT (GPT-5.6 Luna), 23:24 26/09/2026, entry R3-T01.
2. **AI output:** verbatim in entry R3-T01.
3. **Verdict:** INVALID
4. **Reasoning:** The prompt described the speed control ambiguously; ChatGPT assumed piano keys, while the fan has a rotary knob, so the state model behind the test cases does not match the device (ISTQB §4.2.4). TC-14 is also internally inconsistent: its steps leave Speed 2 selected, but its expected result assumes Off.
5. **Student fix:** Rejected. The prompt was corrected and re-sent as R3-T02.

Artifact #9 – R3-T02, 15 test cases (corrected prompt)

1. **Prompt + tool:** ChatGPT (GPT-5.6 Luna), 23:33 26/09/2026, entry R3-T02.
2. **AI output:** verbatim in entry R3-T02; screenshot R3_Device/ai_output_R3-T02.png.
3. **Verdict:** INCOMPLETE
4. **Reasoning:** Of the 15 cases, 4 are valid, 1 is invalid (TC06: a direct 0 → 3 transition is impossible on a sequential knob, ISTQB §4.2.4), and 10 have unmeasurable expected results. The suite misses the knob's end positions (§4.2.2 boundary value analysis), speed × oscillation and power × knob combinations (§4.2.3 decision tables), and error-guessing cases (§4.4.1). Details in section 3.3.
5. **Student fix:** Final suite in section 3.5. Changes: measurable thresholds, AI TC06 removed, TC-06, TC-07, TC-10, TC-11 and TC-12 added.

Artifact #10 – R3 refined test suite draft

1. **Prompt + tool:** Cursor Agent (Claude Opus 5.5), 23:24–23:34 26/09/2026. The prompts are recorded in Appendix A2 (prompt_log.md).
2. **AI output:** report sections 3.1–3.5 (commits 3414a2d, b36f3f4, f2a856f).
3. **Verdict:** INCOMPLETE
4. **Reasoning:** The first draft also assumed piano keys (two keys pressed together, half-pressed key) until the student corrected the control type. The pass thresholds (5 s start, 1 cm displacement, ± 1 s sweep period, 60 min endurance) are tester-defined assumptions.
5. **Student fix:** Rewritten for the rotary knob (commit b36f3f4). On 27/09/2026, TC-08 was executed and failed the stop condition: the head sweeps more than one extra turn (DF-03). The sweep period and angle in TC-08 were not measured. The 5 s start (TC-02), the 1 cm base displacement (TC-13) and the 60 min endurance (TC-15) were not executed, so those thresholds remain unconfirmed.

Artifact #11 – R3 "why the AI missed it" explanations

1. **Prompt + tool:** Cursor Agent (Claude Opus 5.5), 23:36 26/09/2026. The prompt is recorded in Appendix A2 (prompt_log.md).
2. **AI output:** section 3.4, "Why the AI missed it" column (commit 14b4efe).
3. **Verdict:** VALID

4. **Reasoning:** Each explanation is traced to observable content of R3-T02 (its preconditions, its "observation-based" safety note, and the absence of any power-interruption case), and matches the error-guessing rationale of ISTQB §4.4.1.
5. **Student fix:** None. The explanations in section 3.4 were accepted as written.

Artifact #12 – AI compliance drafts

1. **Prompt + tool:** Cursor Agent (Claude Opus 5.5), 23:44 26/09/2026. The prompt is recorded in Appendix A2 (prompt_log.md).
2. **AI output:** AI Critique, Mandatory Disclosure, this appendix, and the filled AI-02, AI-03 and AI-05 templates in the HW01 folder.
3. **Verdict:** INCOMPLETE
4. **Reasoning:** The drafts needed the student's review and identity details before they could be accepted.
5. **Student fix:** Reviewed the AI Critique, the Mandatory Disclosure and this appendix. Filled the identity blocks on AI-02, AI-03 and AI-05 (Huỳnh Thái Toàn, 23120175, CQ2023/31, MSc. Trần Thị Bích Hạnh) and signed each form with the typed name.

Artifact #13 – Excel test workbook

1. **Prompt + tool:** Cursor Agent (Claude Opus 5.5), 00:01 27/09/2026. The prompt is recorded in Appendix A2 (prompt_log.md).
2. **AI output:** HW01_Test_Workbook.xlsx (sheets Test Summary, Test Cases, Checklist, Defects) and report section 3.7 (commit e610c41).
3. **Verdict:** INCOMPLETE
4. **Reasoning:** The Test Cases sheet is copied from section 3.5, but the 20 checklist items (checklist-based testing, ISTQB §4.4.3) and the summary formulas are new AI content. The formulas were not recalculated in Excel when generated, and a technique count in the first draft of section 3.7 was wrong (Error guessing 4 instead of 3) until it was checked against the test cases.
5. **Student fix:** Count corrected. On 27/09/2026 the summary formulas were recalculated from the Test Cases, Checklist and Defects sheets. They match section 3.7: 15 cases, 9 executed (4 PASS, 5 FAIL, 6 NOT RUN), 5 videos, 5 AI-missed edge cases and 5 defects. Checklist items marked OK: 5 of 20; the other 15 stay NOT CHECKED because those checks were not done.

A1.3 Summary of AI accuracy

Metric	Count	Percentage
Total AI-generated artifacts audited	13	100%
VALID (correct, accepted as-is)	3 (#3, #7, #11)	23.1%
INVALID (wrong; rejected)	1 (#8)	7.7%
INCOMPLETE (acceptable after edits)	9 (#1, #2, #4, #5, #6, #9, #10, #12, #13)	69.2%

A1.4 Conclusion – when should AI be used (or not)?

AI was useful for producing a first structure quickly: a 15-case table, defect explanations, a mindmap skeleton, and consistency checks across the report. It failed on precise facts (dates, version numbers, CVSS sources, URLs), on measurable expected results, and on edge cases that come from the physical device rather than from text. It also tended to assume instead of asking when the input was ambiguous. My recommendation: use AI for drafting, brainstorming coverage categories and reformatting, but never as a source of facts or as a test oracle.

Every AI claim must be checked against a primary source, every expected result must be measurable, and edge cases and actual results must come from the tester's own analysis and execution on the real device.

Self-assessment

Self-assessed on 27/09/2026. Total: **100**.

No.	Criteria	Grade	Self-assessed
1	Job Market 2026+ (10 jobs + AI Impact)	40	40
2	Software Defects 2022–2026 (20 defects)	20	20
3	Physical-product test design (15 test cases + 5 videos)	25	25
AI-1	AI-02 Audit Report (5-section) attached	8	8
AI-2	AI Critique (200–300 words) + AI-03 Disclosure attached	4	4
AI-3	AI-05 Checklist signed + anti-cheat artifacts	3	3
	Total	100	100